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FINAL RESEARCH THESIS

ECG Anomaly Detection and Arrhythmia Classification

A Normal-Only Autoencoder with
Temporal-Holdout Evaluation

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Submitted as a final research thesis

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Affidavit

I declare that this thesis is my own work. I have clearly acknowledged all sources of information, software, data, and ideas used in the research. Material quoted or adapted from other work is identified by citation. The experimental results reported in this thesis were obtained from the project implementation described in the document.

Date

Signature

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Matara, August 2026
A.K.S.S. Wimalasena

Abstract

This thesis develops a two-stage electrocardiogram (ECG) system that uses one lead. The first supervised stage decides whether a heartbeat is normal or abnormal. If the beat is abnormal, the second stage classifies it as premature ventricular contraction (PVC), premature atrial contraction (PAC), left bundle branch block (LBBB), right bundle branch block (RBBB), or atrial fibrillation (AFib).

All models receive one ECG channel at a time. A dilated U-Net autoencoder learned normal ECG structure from the complete first stored ECG channel of all 18 MIT-BIH Normal Sinus Rhythm Database records. The WFDB header names this channel ECG1, but it does not state the exact anatomical placement. Fifteen subjects were used for fitting and three different subjects were used for validation. Four reconstruction-error measures were then combined with MLII beat shape, frequency power, and causal RR-interval context. Two separate Extra Trees ensembles performed abnormality detection and abnormal-subtype classification.

The supervised study used every MIT-BIH Arrhythmia Database record that contains MLII: 46 records representing 45 subject clusters. In each record, the first 80% of time was used for development and the final 20% was kept for testing. A 16-beat gap was left before the test boundary to reduce contact between neighbouring windows. Therefore, the test contains later unseen signal from known patients. It is not an unseen-patient test.

On 20,988 temporal-test beats, abnormality accuracy was 98.47%, balanced accuracy was 98.22%, sensitivity was 96.59%, and specificity was 99.85%. Conditional subtype accuracy on 7,964 supported abnormal beats was 95.20%, with macro F1 of 0.9325. On 20,062 supported six-class beats, whole-system accuracy was 96.86% and macro F1 was 0.9303. Equal-subject whole-system accuracy was 96.90%. A 10,000-replicate subject-cluster bootstrap gave a 95% confidence interval of 94.02–99.05%. Its lower bound exceeds 90%, but its 5.03-point width does not meet the planned three-point precision target. Forty-two of 45 subject clusters exceeded 90% ($p = 4.33 \times 10^{-10}$, exact one-sided sign test).

The aggregate target was reached for this known-patient temporal holdout, but not for every subtype. PAC end-to-end recall was 70.05%. The system therefore shows strong performance on later ECG after patient-specific training. However, it does not prove more than 90% accuracy for completely new patients or for every arrhythmia type. It remains a research prototype, not a clinical device.

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1 Introduction

1.1 Introduction

An electrocardiogram (ECG) records the small voltage changes that occur when electrical activity moves through the heart. Electrodes on the body see this activity from different directions. In an ECG trace, the horizontal axis shows time and the vertical axis shows voltage. ECG analysis is useful because the pattern changes when a heartbeat starts in a different place, happens at a different time, or travels through the heart in a different way.

Components of one ECG beat. A normal heartbeat starts when the sinoatrial node creates an impulse. The P wave shows atrial depolarisation. The short, sharp QRS complex mainly shows ventricular depolarisation. The T wave shows ventricular repolarisation. The time between two neighbouring R peaks is called the RR interval. It gives information about heart rate and rhythm regularity [1]. Figure 1.1 shows these parts in a simple way. In real recordings, the same beat can look inverted, stretched, or noisy because of the lead, the patient, the electrodes, and the recording conditions.

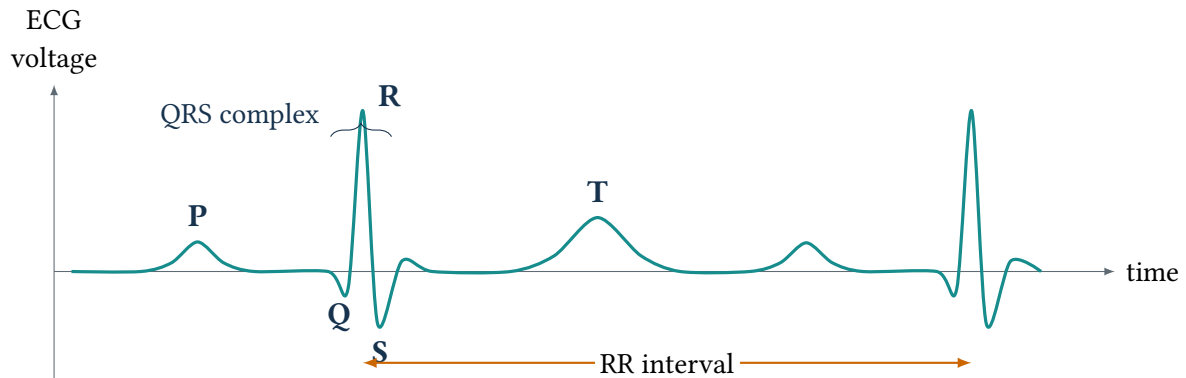


Figure 1.1: Main components of a simplified ECG beat. Shape features describe the P–QRS–T pattern, while RR features describe timing between beats. This is an educational drawing, not a diagnostic template.

Arrhythmia and the classes considered. An arrhythmia is a problem in where the heart's electrical activity starts, when it happens, how it is ordered, or how it travels. It may affect one beat, many repeated beats, or a longer rhythm. This study uses one normal output and five arrhythmia outputs. Table 1.1 summarises the main physiological cause and visible ECG behaviour for each one. These descriptions define the model task only. A clinical diagnosis still needs the full ECG and patient context [2, 3].

Table 1.1: Normal rhythm and the five arrhythmia types considered in this study.

Type	Main reason	Typical ECG behaviour and importance
Normal sinus rhythm (N)	The impulse begins in the sinoatrial node and follows the normal conduction path.	Repeated P–QRS–T patterns and generally regular RR intervals. It supplies the reference against which unusual beats are compared.
PVC	A premature impulse begins in a ventricle instead of following the normal path from the atria.	The beat occurs early and often has a wide, unusual QRS, followed by a pause. Detection is important because it is a clear ventricular abnormality and can occur repeatedly.
PAC	A premature impulse begins in an atrial location outside the sinoatrial node.	The beat occurs early; the P wave may change, while the QRS can remain narrow and similar to normal. Its subtle shape makes PAC difficult to separate from normal beats.
LBBB	Electrical conduction through the left bundle branch is delayed or blocked.	The ventricles activate in an altered sequence, producing a broad, changed QRS. Recognition is important because the pattern can be associated with underlying cardiac disease.
RBBB	Electrical conduction through the right bundle branch is delayed or blocked.	The later part of the QRS changes and can show a second peak in some leads. It must be separated from LBBB because the delayed side is different.
AFib	Atrial electrical activity becomes rapid and disorganised rather than following one regular atrial impulse.	Consistent P waves disappear and RR intervals become irregular. Identification is important because AFib is linked with clinically important complications and requires rhythm-level evidence.

Figure 1.2 shows real labelled ECG segments from the MIT–BIH Arrhythmia Database instead of drawings. Each panel comes from the fixed final 20% of its record. For each stated record and class, the displayed window is the one closest to that held-out group’s mean waveform by squared distance. This rule gives a representative example without choosing the most dramatic-looking beat. The signals were resampled to 128 Hz, standardised, and aligned

so the labelled R peak is at 1.5625 seconds.

Real labeled ECG segments from the locked final 20%

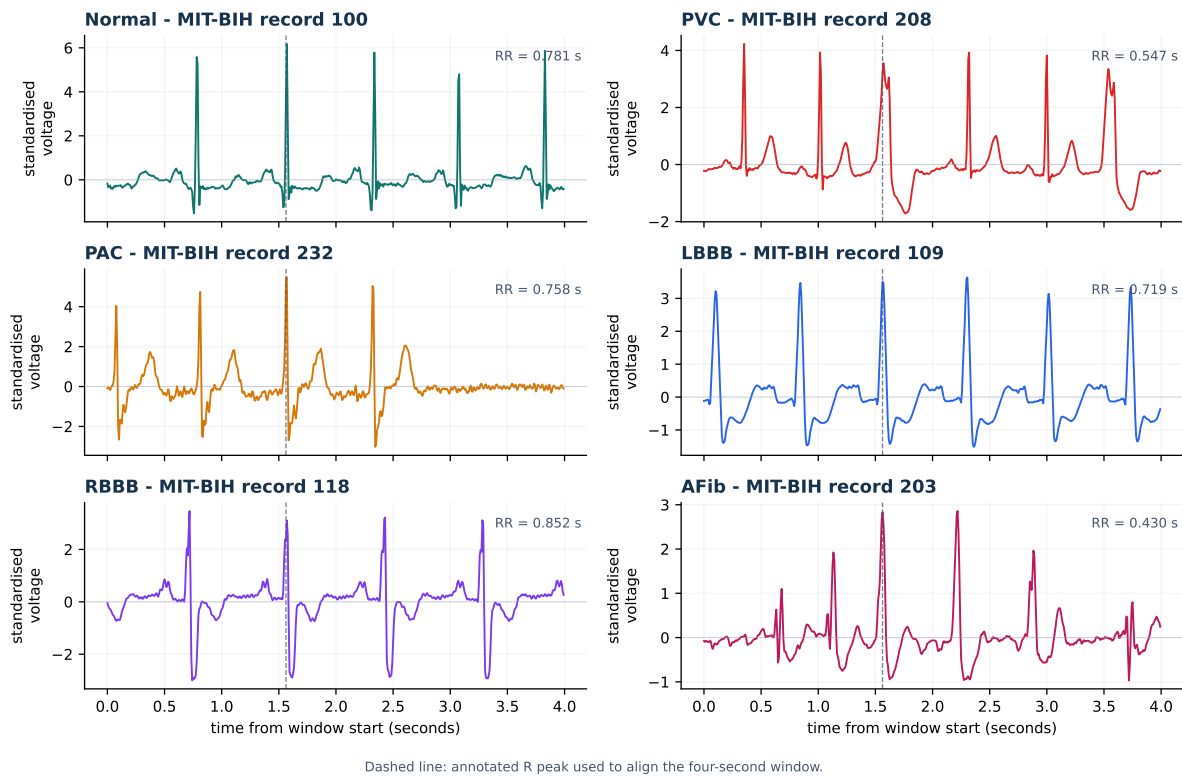


Figure 1.2: Real four-second ECG examples from the MIT-BIH Arrhythmia Database: normal record 100, PVC record 208, PAC record 232, LBBB record 109, RBBB record 118, and AFib record 203. The dashed line marks the annotated central R peak. The examples are used for explanation and are not clinical diagnostic templates.

These examples show why one simple threshold is not enough. PVC, LBBB, and RBBB mainly change beat shape. PAC can occur early while still looking close to a normal beat. AFib is often clearer from changing intervals and rhythm context. For this reason, the final system combines MLII waveform shape, frequency information, RR history, and disagreement with a learned normal reconstruction.

Finding these patterns matters because long ECG recordings can contain too many beats for continuous manual review. An automated research system can mark suspicious regions, count events, and help a reviewer focus attention. At the same time, an error can miss an abnormality or create an unnecessary alarm. Therefore, accuracy, per-class recall, confidence intervals, and clear limits are all needed. The system developed here is for research evaluation only. It is not a replacement for a clinician.

1.2 Research Problem

The central research problem is stated below.

Main research problem: How can an explainable two-stage ECG system use normal-only reconstruction, beat morphology, and heartbeat timing to detect abnormal beats and correctly identify PVC, PAC, LBBB, RBBB, and AFib with more than 90% whole-system accuracy on a fixed temporal test, while providing statistically valid evidence across patients?

The main problem contains four connected sub-problems:

1. ECG shape varies between patients and leads, so harmless personal morphology can resemble an abnormality.
2. A two-stage system can fail at the abnormality gate or at subtype classification; a missed abnormal beat never reaches the second stage.
3. Normal and arrhythmia classes have unequal numbers of beats and patients, so ordinary accuracy can hide failure on a rare class.
4. Thousands of neighbouring beats from one patient are correlated. Testing must keep the final signal unseen and must calculate uncertainty using complete patient clusters rather than treating beats as independent people.

The final experiment handles these problems with a chronological split. The first 64% of each MITDB record trains candidate models, the next 16% selects the models and threshold, and the final 20% is reserved for temporal testing. The fitted classifier does not see the test signal, although it does use earlier ECG from the same patients. The result therefore measures patient-adapted temporal continuation, not performance on completely new patients.

1.3 Objective and Expected Outcomes

Main objective. The main objective is to develop and statistically evaluate an explainable ECG system that first detects abnormal beats and then identifies five supported arrhythmia types in later unseen signal from known patients.

Specific objectives.

1. Train a normal-only autoencoder with the complete first ECG channel from all 18 MIT-BIH Normal Sinus Rhythm Database subjects and use its reconstruction errors as evidence.
2. Use the earlier 80% of every MITDB record containing MLII for supervised development while keeping every final 20% portion unavailable.
3. Build a binary abnormality detector using ECG morphology, spectrum, RR timing, and autoencoder-residual features;

4. Build a second classifier for PVC, PAC, LBBB, RBBB, and AFib using the same MLII signal and causal rhythm context;
5. measure abnormality-detection accuracy, conditional subtype accuracy, and complete six-class system accuracy separately;
6. report the true and correctly classified count for every arrhythmia; and
7. validate uncertainty with equal-subject accuracy, a complete-subject bootstrap confidence interval, and subject-level statistical tests.

Expected outcomes. The planned outcomes are:

- a reproducible, deployable two-stage research prototype;
- abnormality-detection, subtype, and whole-system point accuracies above 90% on the fixed temporal test;
- a whole-system 95% confidence interval whose lower bound exceeds 90% and whose total width is no more than three percentage points;
- per-arrhythmia results that reveal whether every supported class reaches the accuracy target; and
- an honest statement of the data, statistical, hardware, and clinical limits of the model.

These are evaluation targets, not assumed conclusions. Chapter 7 reports which targets were reached and which were not.

2 Literature Review

This chapter reviews the work most relevant to the thesis questions: can a system detect and name ECG abnormalities, and how does performance change when earlier ECG from the same patient is available? The review focuses on four areas: ECG datasets and evaluation protocols, supervised heartbeat classification, normal-only anomaly detection, and statistical validation. It is a focused narrative review, not a systematic review or meta-analysis. Papers were selected because they explain a design choice, show an important alternative, or reveal a risk in patient-independent evaluation.

Headline accuracy is not used to rank the papers. ECG studies often use different leads, labels, record lengths, patient splits, and output units. For example, a patient-specific beat classifier, a 12-lead diagnostic model, and the six-class system in this thesis answer different questions. Their accuracy values cannot be compared directly. This matters because a high score may come from an easier class mapping, or from placing beats from the same person in both training and test data.

PhysioNet provides open physiological databases and common software tools [4]. The MIT-BIH Arrhythmia Database (MITDB) became a widely used benchmark because it contains expert beat annotations and two ECG channels [2, 5]. The MIT-BIH Normal Sinus Rhythm Database provides 18 real long-term recordings from subjects without significant arrhythmias [6]. It is suitable for normal-only learning, but its headers identify the channels only as ECG1 and ECG2. The present study therefore uses one fixed first channel and reports this uncertainty instead of assigning an unsupported anatomical lead name.

The evaluation protocol is as important as the database. A beat-wise random split can place very similar beats from one patient in both training and test sets. An inter-patient split instead tests whether the learned boundary transfers to a different person. De Chazal et al. established an influential inter-patient heartbeat study using ECG morphology and heartbeat-interval features [3]. Later research continued to treat patient variation as a central problem rather than assuming that all beats are independent [7, 8].

This evidence supports a clearly defined evaluation protocol instead of a single ambiguous score. The final experiment keeps chronological order, trains on the first 80% of every record, and tests on the final 20%. This protocol is stricter than a random beat split. However, it measures patient-adapted temporal continuation, not performance on a new patient. The separate NSRDB cohort is used only for the normal autoencoder; no healthy record trains either supervised tree model.

Early automatic ECG systems used measured morphology and timing. De Chazal et al. combined waveform information with RR intervals, showing why beat shape and beat timing should be treated as complementary evidence [3]. This idea remains useful even when the classifier changes: morphology helps describe PVC and bundle-branch beats, while RR history

helps describe premature beats and irregular rhythms.

Huang et al. studied inter-patient separation of normal, LBBB, and RBBB beats with ensemble classifiers [9]. That study is particularly relevant because bundle-branch morphology can persist through a record. However, its three-class task is narrower than the present N/PVC/PAC/LBBB/RBBB/AFib task. Its results therefore justify using morphology and ensemble learning, but they cannot be copied as an expected accuracy for this thesis.

Feature-based models have two practical advantages for this dataset. They make the evidence easier to inspect, and they can be controlled when only a few patients contain a class. Their weakness is that hand-designed summaries may miss useful waveform detail. This trade-off led to the combined design used here: pooled waveform and RR features are kept, while an autoencoder adds learned normal-reconstruction evidence.

One-dimensional convolutional neural networks can learn ECG morphology directly. Kiranyaz et al. developed a patient-specific real-time classifier [10]. Its patient adaptation is useful when labelled beats from the target person exist. It is therefore directly relevant to the final patient-adapted setting, but it does not answer performance before seeing a new patient. Acharya et al. classified arrhythmias from ECG segments with a convolutional network [11]. Segment-level work demonstrates the value of learned temporal features, but its label definition and split must match the target study before accuracy can be compared.

Large deep-learning studies have also reported strong performance on ambulatory single-lead recordings and 12-lead diagnostic ECGs [12, 13]. These works show the potential of deep networks when many patients and labels are available. They also show why dataset scale matters. Their databases, lead systems, and diagnostic outputs are different from the small six-class MITDB experiment, so they are evidence for a method family rather than a performance baseline for this thesis.

Sequence models use neighbouring beats to add rhythm context. Mousavi et al. compared intra-patient and inter-patient heartbeat classification with a sequence-to-sequence approach [7]. Zhang et al. added an adversarial subject classifier so that the learned representation would contain less patient identity [8]. These approaches address generalisation directly, but they also need enough independent patients. The present study has only 45 eligible subject clusters, so the final system uses a simpler tree hierarchy. This is a data-availability decision, not a claim that deep learning is generally inferior.

Autoencoders learn a compressed representation by reconstructing their input [14]. Denoising training corrupts the input but retains a clean target, encouraging the network to learn stable structure rather than an exact copy [15]. In anomaly detection, the model is often trained on normal sequences and reconstruction error is used as evidence that a new sequence differs from the learned normal pattern [16, 17].

Temporal convolutional autoencoders use convolution and dilation to cover longer time patterns efficiently. Thill et al. evaluated this idea on time-series anomaly tasks that included ECG [18]. ECG-specific studies have also investigated convolutional variational and adversarial autoencoders [19, 20]. Together, these studies support using a convolutional autoencoder for normal ECG representation.

The literature does not mean that reconstruction error is a diagnosis. A high-capacity model may reconstruct an abnormal input well, while noise or a different electrode position may

increase the error of a normal input. For that reason, this thesis does not use one reconstruction threshold as the final answer. It uses four residual summaries as features in a supervised abnormal detector. The normal-only idea is kept, but the final decision can combine residual, morphology, spectral, and RR evidence.

ECG datasets contain unequal class counts and unequal numbers of patients per class. A high overall accuracy can therefore hide failure on a rare arrhythmia. Precision–recall analysis is more informative than ROC analysis in strongly imbalanced binary problems [21], while class-balanced learning can increase the influence of rare classes [22]. This thesis therefore reports sensitivity, specificity, balanced accuracy, macro F1, and per-class recall in addition to accuracy.

Thousands of beats from a small number of subjects do not become thousands of independent patients. If individual beats are resampled, the confidence intervals become too narrow. Patient-cluster bootstrap methods keep within-patient dependence [23]. The final experiment gives each of 45 subject clusters equal weight and resamples complete subjects. This also follows modern guidance to report the data split, model development, and validation population clearly [24].

The literature provides strong individual ideas: normal-only reconstruction, learned waveform features, morphology and RR features, patient-invariant learning, specialised bundle-branch classification, and clustered statistical validation. However, reported results usually address only part of the present problem. They may use a different class mapping, adapt to the test patient, use more leads, or evaluate anomaly detection without naming the abnormal type.

The gap addressed by this thesis is therefore not simply the need for another classifier. The need is to evaluate a complete two-stage system while making the generalisation claim explicit:

1. learn normal reconstruction from one fixed channel of all 18 real MIT–BIH normal-sinus recordings;
2. use a chronological 80/20 experiment for patient-adapted temporal evidence;
3. keep each final test portion unavailable during selection; and
4. report separate abnormal, subtype, whole-system, patient-level, and uncertainty results.

Table 2.1 summarises the papers that contributed most directly to the data protocol, model architecture, and evaluation method used in this research.

Table 2.1: Main papers referenced in this study and their contribution.

Paper title	Author(s)	Main contribution used in this research
MIT-BIH Normal Sinus Rhythm Database [6]	George B. Moody	Provides the 18 real long-term recordings without significant arrhythmias used for normal-only autoencoder development; its generic channel names also motivate explicit reporting of lead uncertainty.
Automatic Classification of Heartbeats Using ECG Morphology and Heartbeat Interval Features [3]	Philip de Chazal, Maria O'Dwyer, and Richard B. Reilly	Established the value of combining ECG morphology with RR intervals and carefully defining the evaluation population. This supports the morphology, timing, and reporting choices of the present system.
Reducing the Dimensionality of Data with Neural Networks [14]	Geoffrey E. Hinton and Ruslan R. Salakhutdinov	Provides the representation-learning basis of the autoencoder used to learn normal ECG structure.
Extracting and Composing Robust Features with Denoising Autoencoders [15]	Pascal Vincent, Hugo Larochelle, Yoshua Bengio, and Pierre-Antoine Manzagol	Supports adding noise to the input while keeping a clean reconstruction target, so the autoencoder learns stable normal patterns.
Temporal Convolutional Autoencoder for Unsupervised Anomaly Detection in Time Series [18]	Markus Thill, Wolfgang Konen, and Thomas Baeck	Supports temporal convolution, dilation, and reconstruction error for detecting unusual time-series patterns, including ECG signals.
Inter- and Intra-patient ECG Heartbeat Classification for Arrhythmia Detection: A Sequence to Sequence Deep Learning Approach [7]	Sajad Mousavi and Fatemeh Afghah	Shows the importance of neighbouring-beat rhythm context and clearly separates intra-patient from inter-patient evaluation.
Interpatient ECG Heartbeat Classification with an Adversarial Convolutional Neural Network [8]	Jing Zhang, Aiping Liu, Deng Liang, Xun Chen, and Min Gao	Explains patient-identity leakage and why results from known-patient temporal tests must not be presented as new-patient performance.
Detection of Inter-Patient Left and Right Bundle Branch Block Heartbeats in ECG Using Ensemble Classifiers [9]	Huifang Huang, Jie Liu, Qiang Zhu, Rui Wang, and Guangshu Hu	Supports morphology and ensemble classifiers for LBBB and RBBB, while showing that this specialised three-class task is narrower than the thesis task.
The Precision–Recall Plot Is More Informative than the ROC Plot When Evaluating Binary Classifiers on Imbalanced Datasets [21]	Takaya Saito and Marc Rehmsmeier	Justifies reporting precision, recall, and F1 because the ECG classes are imbalanced and accuracy alone can hide failure.

Paper title	Author(s)	Main contribution used in this research
Bootstrap Estimation of Diagnostic Accuracy with Patient-Clustered Data [23]	Carolyn M. Rutter	Provides the reason for resampling complete patient records instead of treating correlated beats as independent observations.
TRIPOD+AI Statement: Updated Guidance for Reporting Clinical Prediction Models That Use Regression or Machine Learning Methods [24]	Gary S. Collins et al.	Supports transparent reporting of the patient split, model-development process, temporal evaluation, limitations, and intended use.

3 Data

This chapter identifies the ECG sources, explains how the real signals are cleaned, and shows how examples enter model fitting, validation, and the temporal test. Keeping these roles separate prevents healthy data or test labels from being used in the wrong part of the study.

3.1 Data Sources and Roles

The study uses two public PhysioNet databases. The MIT-BIH Normal Sinus Rhythm Database (NSRDB) supplies only healthy ECG for the autoencoder. The MIT-BIH Arrhythmia Database (MITDB) supplies labeled normal and abnormal beats for the detector, subtype classifier, and final temporal evaluation [4, 6, 5].

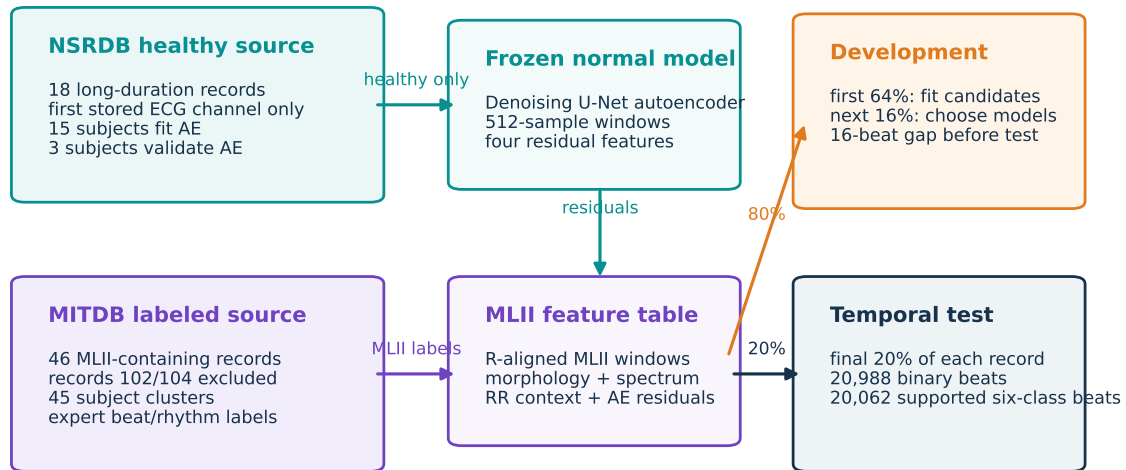
Table 3.1: Data sources and their strictly separated roles.

Source	Available data	Use in this study
MIT-BIH Normal Sinus Rhythm DB (NSRDB)	18 real long-term ECG recordings from five men aged 26–45 and 13 women aged 20–50, with no significant arrhythmias; two stored channels named ECG1 and ECG2; 128 Hz; approximately 23–26 hours per record [6].	The complete first stored ECG channel from every record is processed. Fifteen subjects fit the normal-only autoencoder and three different subjects validate it and determine the normal residual threshold.
MIT-BIH Arrhythmia DB (MITDB)	48 approximately 30-minute, two-channel ambulatory ECG records from 47 subjects (25 men aged 32–89 and 22 women aged 23–89), sampled at 360 Hz, with expert beat and rhythm annotations [5, 2].	Only the 46 records containing MLII are eligible. Their earlier 80% supplies supervised development and their final 20% supplies the temporal test. These records represent 45 subject clusters because records 201 and 202 belong to one person.

Figure 3.1 gives the same information as a flow diagram. In short, NSRDB contributes

only healthy windows for the frozen autoencoder, while MITDB contributes MLII beats for supervised development and the final temporal test.

Dataset roles and evaluation boundaries



The temporal test uses later signal from the same MITDB patients; it is not an unseen-patient test.

Figure 3.1: Dataset roles and evaluation boundaries. The healthy NSRDB channel creates the frozen normal model. MITDB MLII creates the supervised feature table and is split chronologically into development and temporal-test regions.

Why NSRDB is used for healthy training. NSRDB contains real long-duration ECG from subjects reported to have no significant arrhythmias. This matches the autoencoder idea: the network sees normal structure but receives no arrhythmia examples or labels. All 18 subjects and their complete recordings enter the cleaning process, giving 437.49 hours of source ECG. Complete recordings also expose the model to normal within-day variation instead of a short selected excerpt.

Exactly one channel is used. NSRDB headers call the two waveforms ECG1 and ECG2; they do not identify either as a standard anatomical lead. This study therefore selects the first stored channel consistently and describes it as the *first NSRDB ECG channel*. It is not renamed Lead II. The second channel never enters autoencoder fitting, validation, threshold estimation, or residual calculation.

The supervised branch uses MLII only. In the deployed placement, MLII is formed from torso electrodes with the negative electrode near the right upper chest, the positive electrode on the left lower chest or rib region, and a reference electrode on the lower right torso. This modified Lead-II axis usually shows a prominent QRS complex, making beat shape and RR timing clear [5]. The loader selects MLII by its header name rather than assuming a channel number. MLII is stored first in 45 eligible records; record 114 stores V5 first and MLII second.

Records 102 and 104 contain V5 and V2 but no MLII and are excluded. Consequently, the detector, classifier, temporal test, and installed replay all receive one MLII waveform.

The two model stages are one-channel systems, but the channel identities are not falsely treated as identical: the autoencoder learns healthy structure from the first NSRDB channel, while the supervised hierarchy operates on MLII. This source difference is a limitation because the electrode placement for the NSRDB channel is not documented.

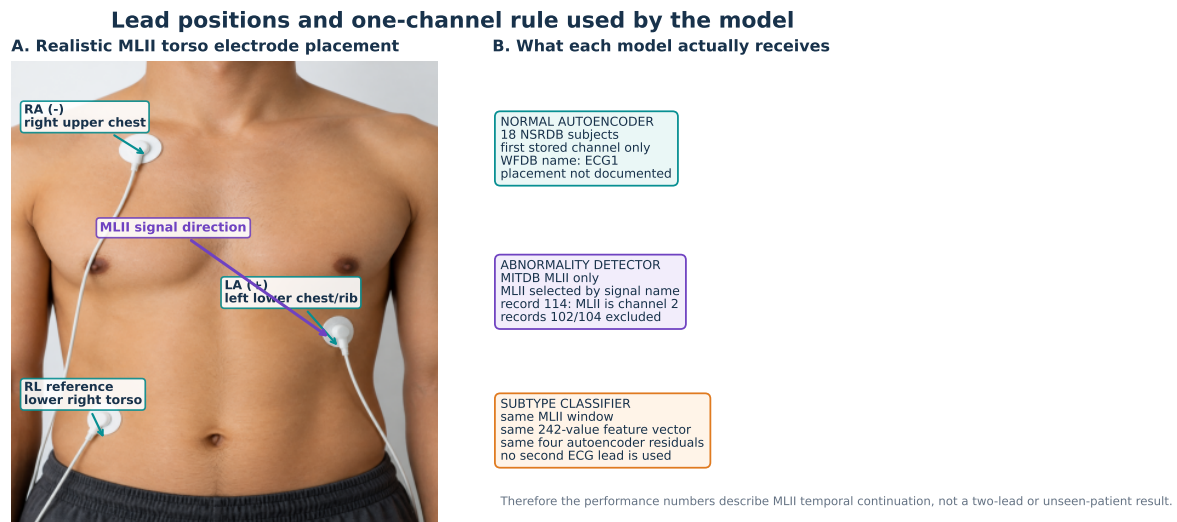


Figure 3.2: Physical MLII placement and the exact one-channel rule. The normal autoencoder uses only the first stored NSRDB ECG channel, whose anatomical placement is unspecified. The supervised and deployed system uses only MLII: negative electrode on the right upper chest, positive electrode on the left lower chest/rib region, and reference electrode on the lower right torso.

Reference labels. MITDB expert annotations are mapped to the six project outputs as shown in Table 3.2. AFib is determined from the rhythm annotation around a beat; the other targets mainly use beat symbols.

Table 3.2: Mapping from MITDB expert annotations to project outputs.

Output	MITDB evidence	Interpretation
N	N, e, j outside AFib/AFL	normal beat
PVC	V, E	ventricular premature or escape beat
PAC	A, a, J, S	supraventricular premature beat
LBBB	L	left bundle branch block beat
RBBB	R	right bundle branch block beat
AFib	normal-like beat inside AFIB/AFL rhythm	atrial fibrillation/flutter context

Unsupported symbols remain abnormal in binary evaluation but are not forced into a wrong subtype. The MLII cache contains 104,847 accepted beats: 63,130 normal and 41,717 abnormal. Of these, 100,159 have a supported six-class label.

3.2 Data Cleaning

Data cleaning converts two different raw ECG sources into the same numerical input: one four-second signal with 512 samples. The cleaning rules are fixed before evaluation and are reused after installation. Figure 3.3 shows the full transformation for one healthy NSRDB segment and one MITDB MLII beat.

3.2.1 Cleaning Checklist

The cleaning pipeline is easier to check as a list. Each item has one purpose and one failure rule.

Table 3.3: Cleaning checklist applied before a signal can become a model input.

Step	What is done	Why it matters
Required channel	NSRDB must provide ECG1; MITDB must provide MLII. Missing required signals stop processing.	Prevents accidental use of ECG2, V5, V2, or another non-matching lead.
Sampling rate	NSRDB remains at 128 Hz. MITDB MLII is resampled from 360 Hz to 128 Hz with polyphase anti-aliasing.	Makes every four-second model window exactly 512 samples.
Band-pass filter	A fourth-order zero-phase Butterworth 0.5–50 Hz filter is applied [25, 26].	Reduces baseline movement and high-frequency noise without shifting the QRS complex.
Amplitude standardisation	Each window is converted to approximately zero mean and unit standard deviation.	Reduces gain differences while preserving the relative P–QRS–T shape.
Window quality check	Windows must be finite, non-flat, and have reasonable variation.	Removes unusable or filter-edge examples before training or scoring.

3.2.2 Window Construction

The autoencoder and supervised hierarchy build windows in different ways, but the final array shape is the same. The autoencoder receives consecutive, non-overlapping four-second windows from the first stored NSRDB channel. The supervised MLII windows are centred on a beat: 200 samples before the R peak and 312 samples after it. During offline experiments, expert annotations give the R-peak location for alignment and later scoring. During live use, detected R peaks replace those annotations.

For samples $x_1, \dots, x_n$ inside one prepared model window, amplitude standardisation is

$$z_i = \frac{x_i - \mu}{\sigma + 10^{-8}}, \quad \mu = \frac{1}{n} \sum_{i=1}^n x_i. \quad (3.1)$$

This operation is applied window by window, not once across an entire record.

3.2.3 Cleaning Yield

The healthy NSRDB recordings produce many candidate windows because complete records are used instead of short excerpts. Across the 18 complete NSRDB records, 393,744 candidate windows were formed. The quality checks kept 345,850 windows and rejected 47,894. These accepted windows were then split into 288,673 autoencoder fitting windows and 57,177 validation windows.

MITDB cleaning produces beat-centered MLII examples. The accepted MLII cache contains 104,847 beats: 63,130 normal and 41,717 abnormal. Unsupported abnormal symbols are still used for binary abnormality evaluation, but they are excluded from six-class subtype scoring.

3.2.4 Installed Use

Installed use applies the same sampling rate, filter, standardisation, 512-sample window length, and feature order. Labels never enter a prediction. The live backend waits until there is enough signal around a detected R peak and enough RR context. It then feeds one prepared MLII window to the frozen system. Neither supervised model is retrained during prediction.

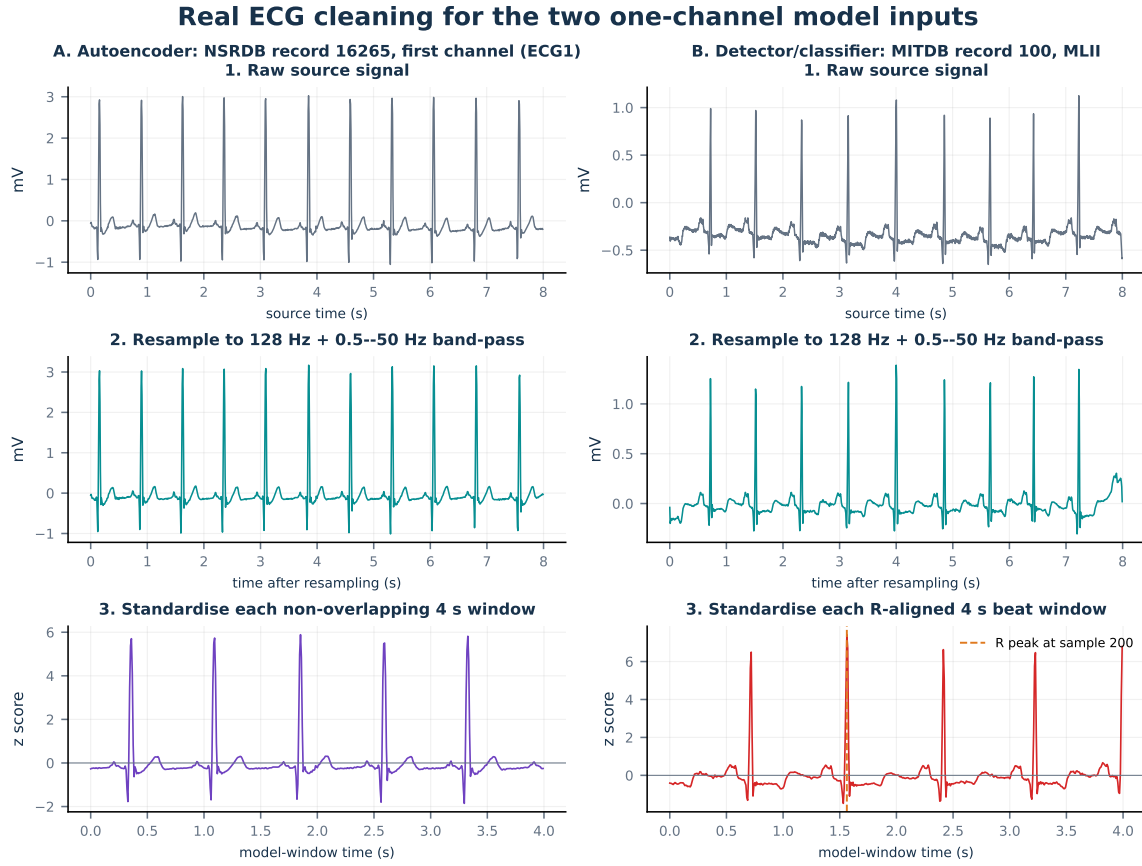


Figure 3.3: Cleaning real ECG for the two roles. The healthy example is the first channel of NSRDB record 16265. The unseen-use example is MLII from MITDB record 100. Both become one 512-sample input, but their documented channel identities remain different.

3.3 Data Preparation

Data preparation decides which cleaned examples are allowed to fit a model, which examples choose model settings, and which examples are held back for the final temporal result. Figure 3.4 shows this flow from raw source to frozen prediction.

3.3.1 Prepared Objects

After cleaning, the study keeps two kinds of prepared object:

- **NSRDB window:** a healthy $[1, 512]$ window from the first stored ECG channel. It has no arrhythmia label and is used only by the normal autoencoder.
- **MITDB beat:** an R-aligned MLII $[1, 512]$ window with record identity, binary normal/abnormal truth, supported subtype truth when available, and RR timing context.

The autoencoder input tensor has shape $[B, 1, 512]$. The supervised detector and subtype

classifier do not receive raw samples alone. Each MITDB beat is converted into the same 242-value MLII feature vector:

$$\begin{aligned} 242 \text{ values} &= 230 \text{ morphology, spectrum, statistics, and base-RR values} \\ &+ 8 \text{ causal RR-history values} \\ &+ 4 \text{ autoencoder residual values.} \end{aligned}$$

The subtype model is fitted only on the five supported abnormal classes: PVC, PAC, LBBB, RBBB, and AFib.

3.3.2 Splits

The autoencoder split is subject-disjoint. A seeded split assigns 15 complete NSRDB records to fitting and three different records (16420, 16483, and 16795) to validation. This produces 288,673 fitting windows and 57,177 validation windows. Validation selects the checkpoint and estimates the normal residual threshold. There is no separate healthy test because the autoencoder is frozen before it contributes residual features to MITDB evaluation.

The MITDB split is chronological inside every eligible record:

- the first 64% fits candidate supervised models;
- the next 16% selects the model structures and abnormal-probability threshold;
- a 16-beat gap is removed before the test boundary; and
- the final 20% evaluates the completed frozen hierarchy.

This is a temporal continuation test from known patients. It is stronger than a shuffled-beat split, but it is not an unseen-patient test.

3.3.3 Exact Model Allocation

Table 3.4: Training, validation, and temporal-test data supplied to each model.

Model	Candidate training	Validation	Final fitting and testing
Autoencoder	288,673 first-channel windows from 15 complete NSRDB records	57,177 windows from 3 different complete NSRDB records	Frozen while every MLII test beat produces four residual features
Abnormality detector	66,344 MLII beats: 40,505 normal and 25,839 abnormal	16,043 beats: 9,671 normal and 6,372 abnormal	Fitted on 83,123 development beats; tested on 20,988 temporal-test beats
Subtype classifier	22,864 supported abnormal MLII beats	5,657 supported abnormal beats	Fitted on 28,784 supported abnormal beats; conditionally tested on 7,964 supported abnormal beats

The supported six-output test contains 20,062 beats: 12,098 N, 1,527 PVC, 591 PAC, 1,482 LBBB, 1,383 RBBB, and 2,981 AFib. Another 926 unsupported abnormal beats remain in binary testing but not six-class scoring.

Data preparation during training and during unseen use

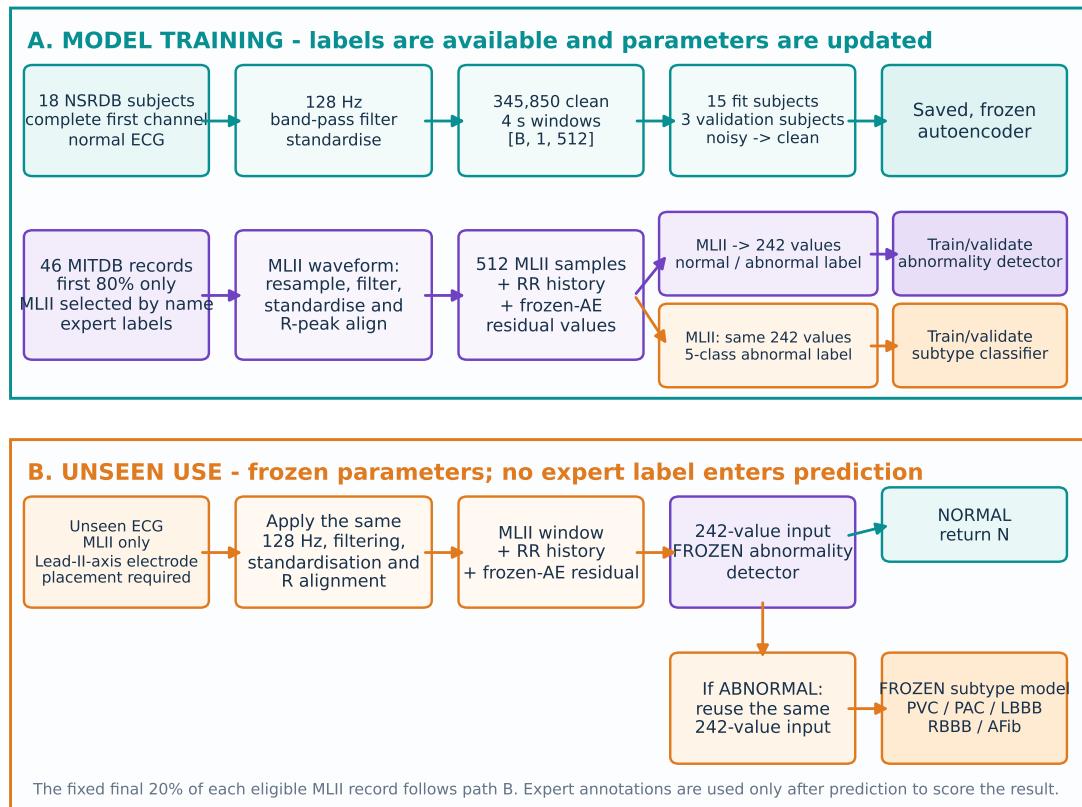


Figure 3.4: Data preparation during model development and use. Panel A separates normal-only autoencoder development from labeled MLII development. Panel B shows one new MLII signal passing through the frozen abnormality gate and then, only when abnormal, the subtype classifier.

3.3.4 Feeding the Frozen System

During autoencoder fitting, Gaussian noise with standard deviation 0.03 is added to each input while the clean NSRDB window remains the target. During supervised fitting, record- and class weights reduce domination by long records and common classes. During prediction, one MLII window and its causal RR history are converted into the same 242-value vector used in the temporal experiment. A normal decision stops. An abnormal decision reuses that vector for PVC, PAC, LBBB, RBBB, or AFib classification.

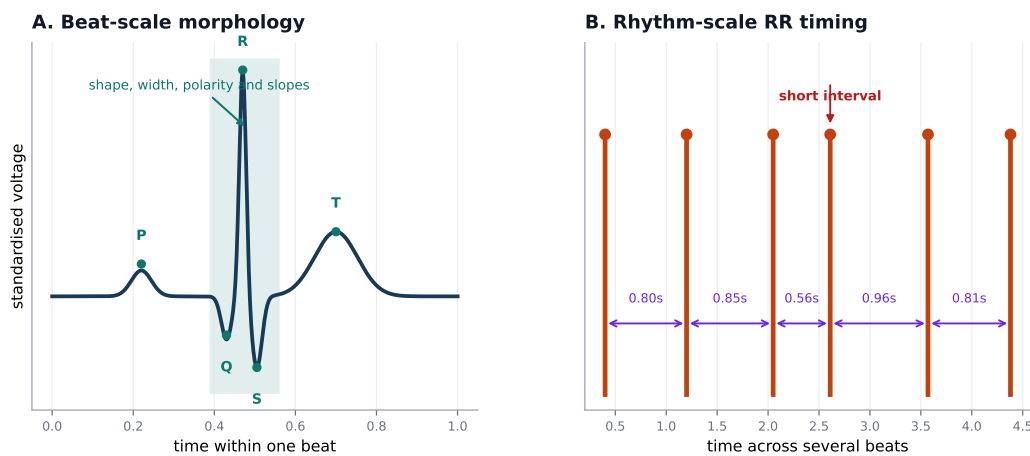
4 Theoretical Foundations

This chapter explains the theory behind the final ECG system. It follows the actual implementation: a one-lead MLII beat classifier that uses waveform shape, RR timing, autoencoder residuals, and two supervised Extra Trees ensembles. The chapter does not assume a second ECG lead or an offline record-summary input, because the deployed model does not use those inputs.

4.1 Why ECG evidence has more than one form

An ECG is a voltage signal over time. For arrhythmia classification, useful evidence appears at two related scales. The first is beat-scale morphology: what the P–QRS–T shape looks like around one R peak. The second is rhythm timing: when the beat occurs relative to neighbouring beats. The final model uses both because they answer different questions.

Two ECG evidence scales used by the final model



The final feature vector joins these scales with four normal-reconstruction residuals.

Figure 4.1: Two evidence scales used by the final model. Morphology describes the local shape of a beat; RR timing describes rhythm context around the beat.

PVC, LBBB, and RBBB can strongly affect QRS shape, width, slopes, and polarity. PAC and AFib may be easier to detect when timing evidence is also present: premature beats create short RR intervals, pauses create long RR intervals, and AFib creates repeated irregularity. Published heartbeat-classification work also shows the value of combining waveform morphology with heartbeat-interval features [3, 7].

4.2 Feature extraction theory

The raw 512-sample MLII window contains more detail than the final tree models need. Small changes in R-peak alignment or noise can change many raw sample values. The system therefore converts the window into compact measurements that keep useful ECG evidence while reducing unnecessary sample-level detail.

Transparent operations that create the final feature vector

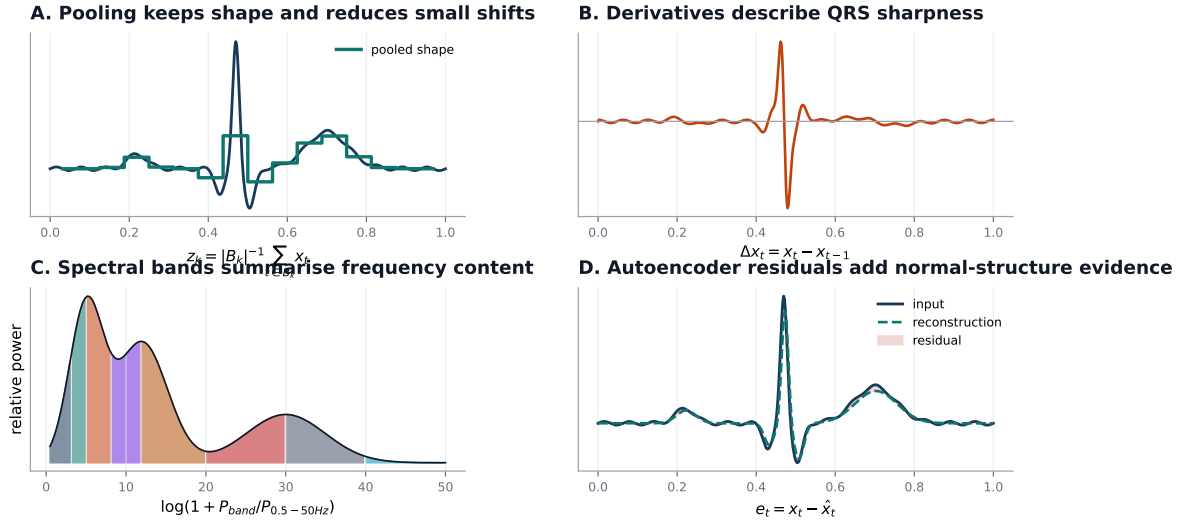


Figure 4.2: Feature-extraction operations behind the final 242-value vector. The tree models learn from compact numerical evidence, not from raw ECG samples directly.

For a pooled region B_k , the pooled value is

$$z_k = \frac{1}{|B_k|} \sum_{t \in B_k} x_t. \quad (4.1)$$

Pooling keeps the average shape in a small region and makes the representation less sensitive to a one- or two-sample shift. The final system uses 64 whole-window pooled values and 80 central morphology pooled values.

The derivative

$$\Delta x_t = x_t - x_{t-1} \quad (4.2)$$

describes how quickly voltage changes. This matters because QRS abnormalities often change slopes and sharpness. The system keeps 60 pooled derivative values from the central beat region.

Frequency features describe how signal energy is spread from 0.5 to 50 Hz. For a frequency band b , the relative band feature is

$$s_b = \log \left(1 + \frac{P_b}{P_{0.5-50}} \right), \quad (4.3)$$

where P_b is the power inside the band and $P_{0.5-50}$ is total power in the ECG band used by the feature extractor. This band-limited view follows the general ECG idea of keeping physiological content while reducing drift and high-frequency interference [1, 25].

Eleven summary statistics add scale, range, energy, skewness, and kurtosis. Together, the waveform, derivative, spectral, and statistical values form 223 non-RR, non-autoencoder features.

4.3 RR timing theory

Let r_i be the time of the R peak of beat i . The RR interval is

$$RR_i = r_i - r_{i-1}. \quad (4.4)$$

RR values describe rhythm. A short interval can suggest a premature beat. A long interval can suggest a pause. Repeated interval changes support rhythm irregularity.

The final live system classifies a beat only after it has enough context to compute the same RR information used in the final experiment. It uses seven base RR values, including previous RR, next RR, local mean RR, interval ratios, local coefficient of variation, and heart rate. It then appends eight history-based values from already completed intervals: median, interquartile range, median absolute deviation, RMSSD, pNN50, relative range, trend, and latest interval change.

RMSSD is one example:

$$\text{RMSSD} = \sqrt{\frac{1}{K-1} \sum_{j=2}^K (RR_j - RR_{j-1})^2}. \quad (4.5)$$

The next-RR requirement is important. It creates a small beat-level delay during streaming, but it also makes the live RR feature vector match the experiment.

4.4 Normal-only reconstruction theory

An autoencoder contains an encoder f_θ and decoder g_ϕ . It is trained to reconstruct a normal ECG window x :

$$\hat{x} = g_\phi(f_\theta(x)), \quad \min_{\theta, \phi} \frac{1}{T} \sum_{t=1}^T (x_t - \hat{x}_t)^2. \quad (4.6)$$

This follows the representation-learning idea of autoencoders [14]. The final autoencoder is trained with a denoising objective: small Gaussian noise is added to the input while the target remains the clean normal window. Denoising encourages stable structure instead of exact noise copying [15].

Normal-only reconstruction: residual is evidence, not diagnosis

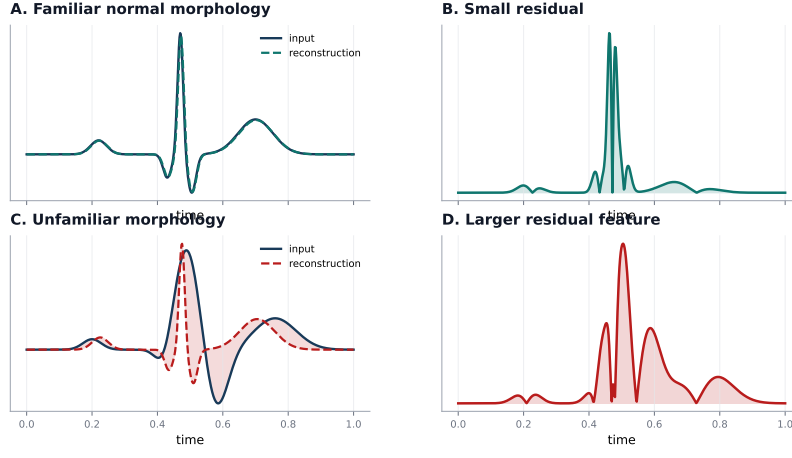


Figure 4.3: Normal-only reconstruction principle. A larger residual can indicate novelty, but it is evidence only; the final arrhythmia decision is made by the supervised hierarchy.

If the autoencoder has learned common normal structure, an unfamiliar or abnormal beat may be reconstructed less accurately. This is the one-class anomaly detection idea [16]. It is not a guarantee. A flexible decoder may reconstruct an abnormal waveform well, and noise may create a large residual for a normal waveform. Therefore, the final system uses autoencoder mismatch as four supervised input features, not as the final diagnosis rule.

Let $e_t = x_t - \hat{x}_t$ be the residual. The four residual features are:

$$e_{\text{MAE}} = \frac{1}{T} \sum_t |e_t|, \quad (4.7)$$

$$e_{\text{MSE}} = \frac{1}{T} \sum_t e_t^2, \quad (4.8)$$

$$e_{\text{local}} = \frac{1}{|Q|} \sum_{t \in Q} |e_t|, \quad (4.9)$$

$$e_{\Delta} = \frac{1}{T-1} \sum_{t=2}^T |e_t - e_{t-1}|. \quad (4.10)$$

They measure whole-window mismatch, squared mismatch, central morphology mismatch, and residual sharpness. The final architecture appends these four values to the MLII waveform and RR features.

4.5 Hierarchical classification theory

The final problem is not solved with one direct classifier. The system first asks whether the beat is abnormal. Only when this binary gate is true does it ask which supported abnormal subtype is most likely.

Hierarchical probability logic in the final model

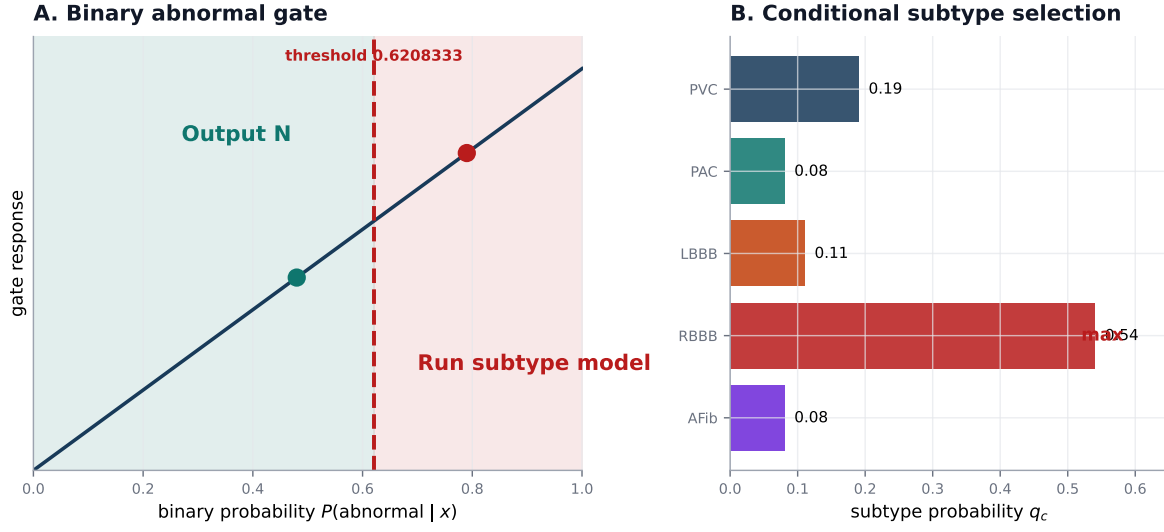


Figure 4.4: Probability logic of the final hierarchy. The subtype model is conditional: it is used only after the abnormality gate is passed.

For beat i , the binary detector estimates $p_i = P(\text{abnormal} | x_i)$. The final threshold rule is

$$a_i = \mathbb{I}[p_i > 0.6208333]. \quad (4.11)$$

If $a_i = 0$, the output is N. If $a_i = 1$, the subtype model selects the highest probability among PVC, PAC, LBBB, RBBB, and AFib:

$$\hat{y}_i = 1 + \arg \max_{c \in \{1, \dots, 5\}} q_{ic}. \quad (4.12)$$

This hierarchy has a practical advantage. Unsupported abnormal MIT-BIH symbols can still contribute to binary abnormality testing without being forced into one of the five supported subtype labels.

4.6 Why Extra Trees fits this feature design

The final 242-value vector mixes pooled waveform values, derivatives, spectral power, signal statistics, RR timing, and autoencoder residuals. Their relationship to class labels is non-linear. Extra Trees fits this setting by building many randomised decision trees and averaging their probability outputs.

At a decision-tree node with class probabilities p_c , entropy is

$$H = - \sum_c p_c \log_2 p_c. \quad (4.13)$$

A split is useful when it reduces the weighted child entropy. Extra Trees adds randomness by selecting random feature subsets and random split candidates. Averaging many such trees reduces the instability of a single tree while retaining non-linear interactions between features.

This theory also explains why Extra Trees is suitable for the final data scale. MITDB contributes many beats but only 45 subject clusters. A very high-capacity end-to-end neural classifier could learn patient identity instead of robust ECG rules [8]. The Extra Trees hierarchy is still high-capacity, but it uses transparent tabular evidence and is easier to audit against the final temporal protocol. This does not mean that tree ensembles are always better than deep learning.

4.7 Class and patient imbalance theory

ECG beat datasets are imbalanced in two ways. Some classes are rare, and some patients contribute many more beats than others. Without weighting, a model can mostly learn common normal beats or a few large records.

The final training code uses record-aware class weights. For class c in record r , the weight is proportional to

$$w_{cr} = \frac{1}{n_c} \left(\frac{n_c}{R_c n_{cr}} \right)^\alpha, \quad (4.14)$$

where n_c is total support for class c , R_c is the number of records containing class c , and n_{cr} is the number of class- c beats in record r . The first factor supports class balancing; the second factor reduces domination by a single record. The exponent α controls how strongly record balancing is applied and is selected in validation. Class-balanced learning is important when rare classes would otherwise have little influence [22].

4.8 Temporal patient adaptation theory

ECG morphology contains patient-specific information such as body geometry, electrode placement, baseline QRS shape, conduction pattern, and background rhythm. If earlier labelled ECG from a patient is available during development, the model can adapt to that patient's baseline. Later ECG from the same patient is then easier than ECG from a completely unseen patient.

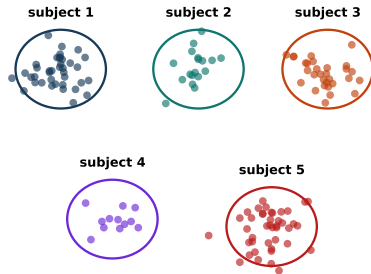
This is the theory behind the final temporal-holdout design. It is useful for monitoring after a labelled calibration period, but it changes the claim. The final result is a known-patient temporal-continuation result. It should not be reported as unseen-patient generalisation unless a patient-disjoint experiment is run. The 16-beat boundary gap reduces direct overlap at the split boundary, but it does not remove shared patient identity.

4.9 Validation theory

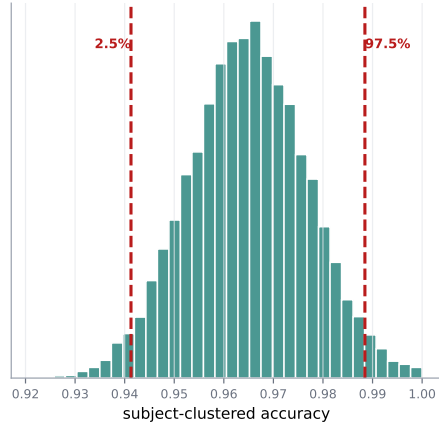
Beat-level accuracy alone is not enough. If one patient contributes 2,000 beats, those beats are related. They are not equivalent to 2,000 independent patients. The final validation therefore reports beat metrics and also uses subject-clustered uncertainty.

Why validation must respect subject clusters

A. Many beats, but fewer independent subjects



B. Bootstrap resamples subjects, not individual beats



This supports a known-patient temporal-continuation claim, not an unseen-patient claim.

Figure 4.5: Theoretical reason for subject-clustered validation. Clustered resampling prevents one long record from being treated as many independent people.

Balanced accuracy gives equal weight to class recalls, so normal beats cannot hide poor abnormal recall. Macro F1 gives each class equal importance, so a failure on a rare class is visible. Precision–recall style measures are also important under class imbalance [21]. For confidence intervals, the final experiment resamples complete subject clusters rather than individual beats, following the general diagnostic-statistics principle that clustered data need clustered analysis [23].

4.10 Theory-to-system alignment

Table 4.1: How the theory maps to the implemented final system.

Theory concept	Implemented in the final system
Beat-scale morphology	512-sample MLII window with pooled full-window, central, and derivative features
Rhythm timing	15 RR features; live scoring waits until next-RR context is available
Normal-only reconstruction	frozen NSRDB autoencoder supplies four residual features
Hierarchical probability	240-tree binary gate first; 260-tree subtype model only for gated abnormal beats
Class/patient imbalance	record-aware class weighting with validation-selected record-power parameter
Temporal adaptation	earlier MLII timeline trains/selects the model; final 20% timeline is tested
Subject-aware uncertainty	subject-clustered bootstrap over 45 clusters, with records 201 and 202 treated as one cluster

5 Final Model Architecture

This chapter explains the final system saved and used by the application. The installed model is a one-lead MLII beat classifier. It does not use two ECG leads, does not retrain during live use, and does not make an arrhythmia decision directly from the autoencoder threshold. The final decision is made by two supervised Extra Trees models. Both receive the same 242-value feature vector.

5.1 Architecture in one sentence

For each detected MLII heartbeat, the system takes a 512-sample window aligned to the R peak, builds waveform, rhythm, and normal-reconstruction features, sends those features to a binary abnormality gate, and sends only gated abnormal beats to a five-class subtype classifier.

Final ECG system: from one MLII beat to one decision

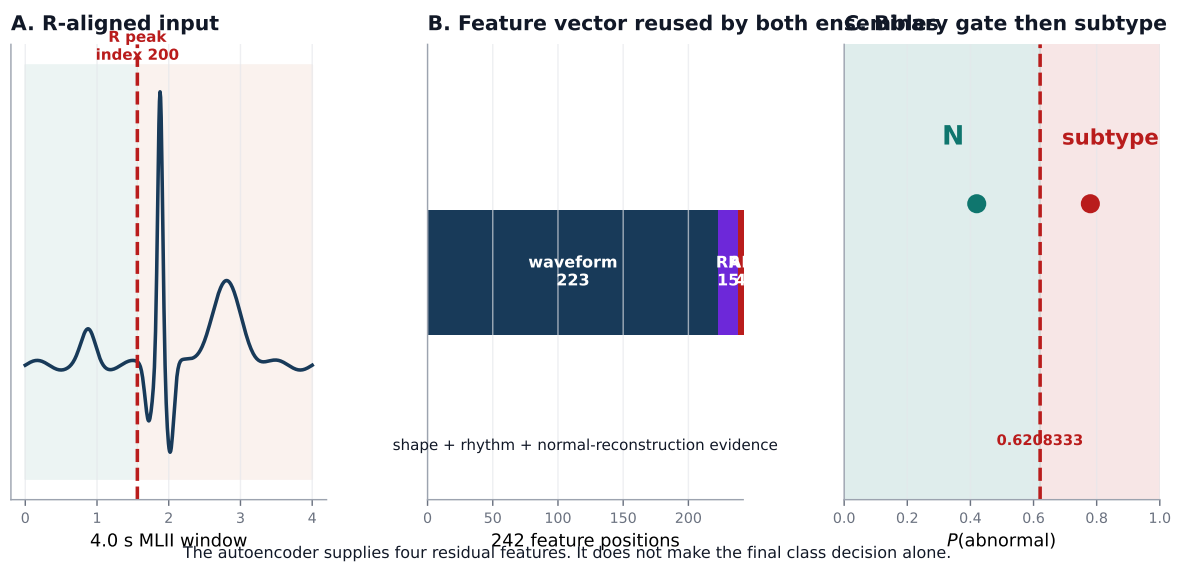


Figure 5.1: Exact runtime order of the final ECG system. The autoencoder supplies features; the supervised tree models make the final class decision.

5.2 Input signal and beat window

The deployed system accepts one ECG channel. For database replay and final testing that channel is MIT-BIH MLII, selected by signal name. Records 102 and 104 are excluded

because they do not contain MLII. For sensor sessions, the API requires the user to declare `lead_name='MLII'` and rejects other declared leads. This rule prevents the model from silently scoring a lead that does not match the supervised training signal.

The classifier is beat based, not arbitrary-window based. The live backend collects MLII samples, detects R peaks, and waits until enough rhythm context is available. For a scorable beat, it extracts a fixed 512-sample window at 128 Hz:

- 200 samples before the R peak;
- 312 samples after the R peak;
- total duration of 4.0 seconds.

This same window geometry is used in the final temporal-holdout experiment and in the installed session pipeline.

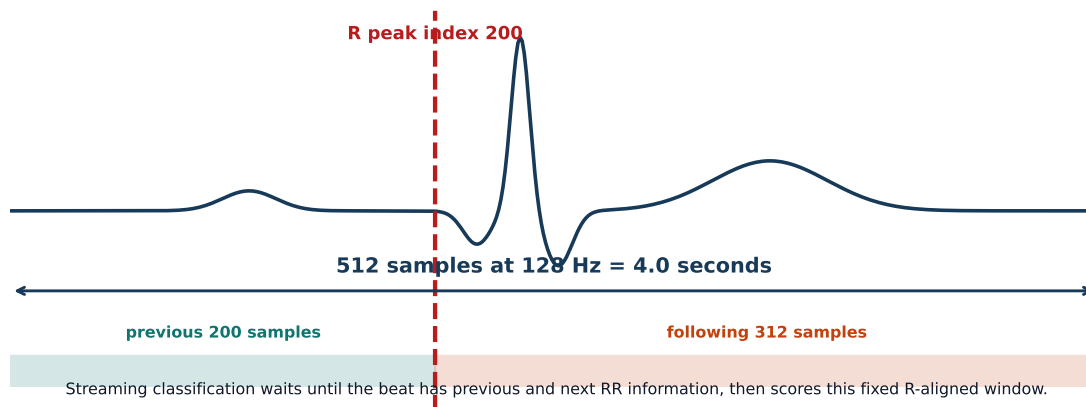


Figure 5.2: R-peak-aligned beat window used by the final model. The R peak is at sample index 200, giving the model pre-QRS context and post-QRS recovery information.

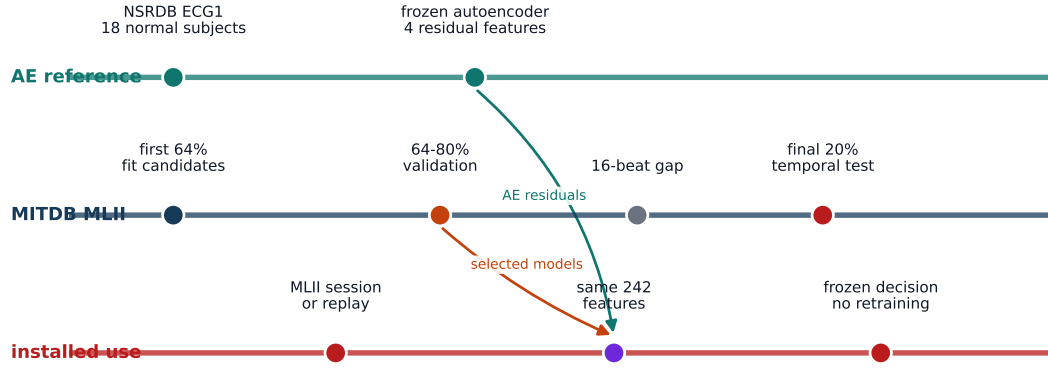
5.3 Three trained components

The final model bundle contains three trained components:

1. a frozen normal-only autoencoder;
2. a binary Extra Trees abnormality detector;
3. a five-class Extra Trees abnormal-subtype classifier.

The autoencoder is trained first and then frozen. The two supervised tree models are then fitted on MLII features from the temporal development portion of MIT-BIH Arrhythmia Database records.

Training, testing, and installed use of the final bundle



The performance claim is known-patient temporal continuation: later MLII from patients whose earlier MLII was used for development.

Figure 5.3: Separation between normal-reference training, supervised temporal development, final testing, and installed use. Live/session use applies the frozen bundle; it does not update model weights.

5.4 Normal-only autoencoder

5.4.1 Purpose

The autoencoder is a normal-reconstruction branch. It learns how normal ECG windows can be reconstructed. The final system then uses reconstruction error as extra evidence. The autoencoder does not decide “normal” or “arrhythmia” by itself in the final classifier.

For an input window $x \in \mathbb{R}^{512}$, the autoencoder output is

$$\hat{x} = g_{\phi}(f_{\theta}(x)), \quad (5.1)$$

where f_{θ} is the encoder and g_{ϕ} is the decoder. Training uses a denoising objective: Gaussian noise with standard deviation 0.03 is added to the input, but the target remains the clean standardised window. The training loss is

$$\mathcal{L}_{AE} = \frac{1}{512} \sum_{t=1}^{512} (x_t - \hat{x}_t)^2. \quad (5.2)$$

This objective encourages stable normal morphology instead of copying small noise patterns [15].

5.4.2 Network structure

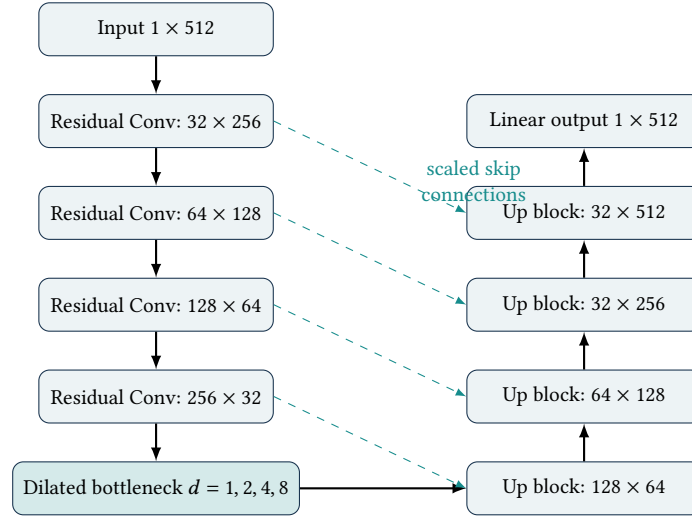


Figure 5.4: Layer flow of the 2,680,641-parameter dilated U-Net autoencoder. The encoder compresses the window, the dilated bottleneck expands temporal context, and the decoder reconstructs the original 512-sample signal.

Each residual encoder block contains two kernel-seven convolutions, Group Normalisation, LeakyReLU, dropout, and a residual projection. Four kernel-three bottleneck blocks use dilation 1, 2, 4, and 8. Dilation increases the receptive field without deleting time samples. The decoder uses transposed convolutions and skip connections scaled by 0.5. The final layer is linear because the standardised ECG signal contains positive and negative values.

5.4.3 Normal-reference data and checkpoint

The autoencoder checkpoint was developed from complete recordings of all 18 subjects in the MIT–BIH Normal Sinus Rhythm Database. Fifteen subjects supply 288,673 fitting windows and three different subjects supply 57,177 validation windows. The selected NSRDB header channel is ECG1. The database does not document this channel as anatomical Lead II, so the report treats it as a normal ECG reference channel rather than renaming it MLII [6].

The final checkpoint uses Adam, learning rate 0.001, batch size 256, and two epochs. Validation MSE changed from 0.061195 after epoch 1 to the best saved value, 0.061193, after epoch 2. The autoencoder artifact also stores a normal-residual threshold of 0.158893, but the final hierarchy does not use that threshold as the final arrhythmia rule.

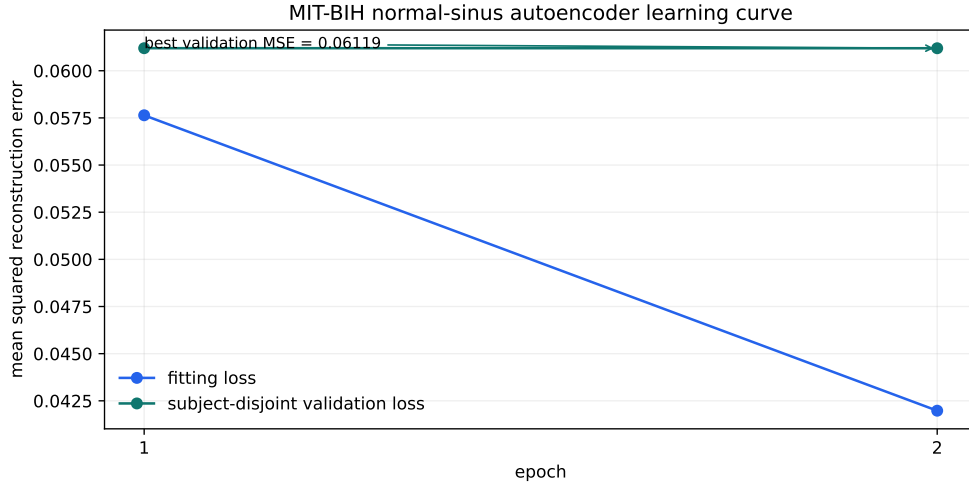


Figure 5.5: Autoencoder fitting and subject-disjoint validation loss. The checkpoint with the lowest validation loss is retained before the supervised MLII stages are trained.

5.4.4 Four autoencoder residual features

Let $e_t = x_t - \hat{x}_t$ be the reconstruction residual. The final feature system uses four scalar residual summaries:

$$f_1 = \frac{1}{512} \sum_t |e_t| \quad \text{whole-window MAE,} \quad (5.3)$$

$$f_2 = \frac{1}{512} \sum_t e_t^2 \quad \text{whole-window MSE,} \quad (5.4)$$

$$f_3 = \frac{1}{240} \sum_{t=104}^{343} |e_t| \quad \text{central morphology MAE,} \quad (5.5)$$

$$f_4 = \frac{1}{511} \sum_{t=2}^{512} |e_t - e_{t-1}| \quad \text{residual-change MAE.} \quad (5.6)$$

The central feature focuses on the QRS and nearby morphology. The residual-change feature becomes large when the reconstruction misses sharp slopes. These four values are appended to the transparent MLII and RR features.

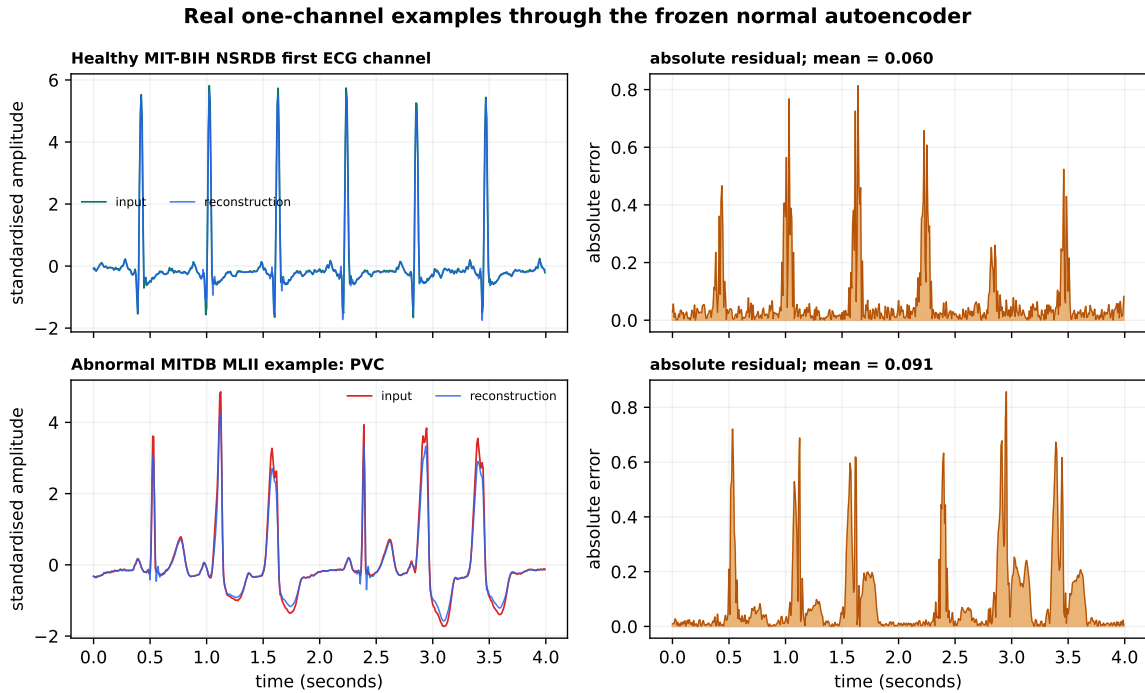


Figure 5.6: Real healthy and abnormal ECG examples through the frozen autoencoder. The residual between input and reconstruction is converted into four numerical features for the supervised detector.

5.5 Final 242-value feature vector

The saved final bundle declares 242 total features, 15 RR features, required autoencoder features, and four autoencoder residual values. Therefore every supervised prediction uses:

$$\begin{aligned}
 &223 \text{ waveform/spectrum/statistics values} \\
 &\quad + 15 \text{ RR timing values} \\
 &\quad + 4 \text{ autoencoder residual values} = 242.
 \end{aligned}
 \tag{5.7}$$

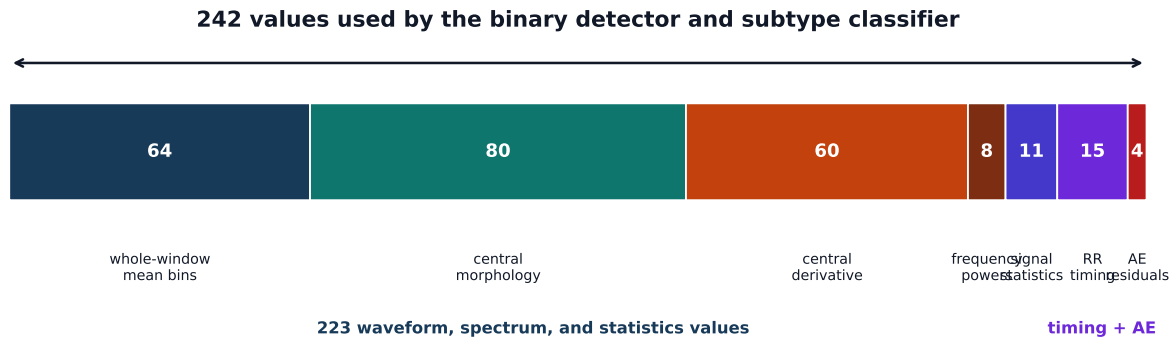


Figure 5.7: Feature-vector layout used by both supervised models. The same 242-value vector is passed first to the abnormality detector and, if required, to the subtype classifier.

Table 5.1: Organised feature groups in the final architecture.

Group	Values	What it tells the model
Whole-window mean bins	64	broad four-second MLII waveform shape
Central morphology bins	80	P–QRS–T shape around the R peak
Central derivative bins	60	sharp slopes and rapid QRS changes
Frequency-band powers	8	relative energy from 0.5 to 50 Hz
Signal statistics	11	scale, range, energy, skewness, and kurtosis
RR timing context	15	previous/next interval, local rate, variability, median, interquartile range, MAD, RMSSD, pNN50, range ratio, trend, and latest RR change
Autoencoder residuals	4	how strongly the beat disagrees with the learned normal-reconstruction pattern

The final experiment constructs the same 242 values in two steps: first 230 values are built from 223 MLII waveform/statistical values plus seven base RR values; then eight causal RR-history values and four autoencoder residual values are appended. During live/session prediction, the backend builds the same final vector directly from 223 MLII waveform/statistical values, 15 RR values, and four autoencoder residual values.

5.6 Binary abnormality detector

The first supervised model is the abnormality detector. It receives all 242 features and estimates $p_i = P(\text{abnormal} \mid x_i)$ for beat i . The selected detector is an Extra Trees classifier with:

- 240 trees;
- entropy split criterion;

- fully grown trees;
- minimum leaf size of one sample;
- square-root feature sampling at each split;
- record/class weighting with record-power $\alpha = 0.35$.

The decision threshold was selected on the 64–80% validation region of each development record:

$$a_i = \mathbb{I}[p_i > 0.6208333]. \quad (5.8)$$

If $a_i = 0$, the final output is normal class N. No subtype classifier is used for that beat.

For tree b , let $p_b(x_i)$ be the abnormal class proportion in the reached leaf. The ensemble probability is

$$p_i = \frac{1}{240} \sum_{b=1}^{240} p_b(x_i). \quad (5.9)$$

Averaging many randomly split trees reduces the instability of one decision tree while still modelling non-linear links among ECG shape, RR timing, and autoencoder residuals.

5.7 Abnormal-subtype classifier

The second supervised model is used only after the binary gate marks a beat as abnormal. It is fitted only on supported abnormal classes 1–5, not on N and not on unsupported MIT–BIH symbols. The selected subtype model is an Extra Trees classifier with:

- 260 trees;
- entropy split criterion;
- fully grown trees;
- minimum leaf size of one sample;
- one-third feature sampling at each split;
- record/class weighting with record-power $\alpha = 0.25$.

For an abnormal beat, the subtype classifier returns probabilities over the five supported abnormal outputs: PVC, PAC, LBBB, RBBB, and AFib. The final subtype is

$$\hat{y}_i = 1 + \arg \max_{c \in \{1, \dots, 5\}} q_{ic}. \quad (5.10)$$

The final saved bundle sets `reject_low_confidence=False`. Therefore the evaluated and installed final pipeline do not add a post-hoc “Unclassified abnormal” rejection step; every gated abnormal beat receives the subtype with the largest subtype probability.

5.8 Final decision logic

The complete prediction rule is:

1. Validate that the input is a 512-sample MLII beat window and that 15 RR features are available.
2. Reconstruct the same window with the frozen autoencoder.
3. Build the 242-value vector from MLII shape, RR timing, and four autoencoder residual summaries.
4. Estimate $p_i = P(\text{abnormal} \mid x_i)$ using the 240-tree binary detector.
5. If $p_i \leq 0.6208333$, output N with confidence $1 - p_i$.
6. If $p_i > 0.6208333$, run the 260-tree subtype model and output the highest-probability supported subtype.

This is the same hierarchy used in the final temporal-holdout evaluation. The model does not use test labels while predicting. The labels are used only after prediction to compute accuracy, F1, confusion matrices, and subject-clustered uncertainty.

5.9 Why this architecture was selected

Table 5.2: Reason for each major final-architecture decision.

Decision	Reason
MLII-only supervised input	matches the signal available in 46 eligible MIT-BIH records and avoids hidden two-lead assumptions
R-peak alignment	puts comparable cardiac events in comparable positions inside every 512-sample window
Waveform and derivative bins	preserve beat morphology, including sharp QRS changes
Frequency powers and statistics	add compact information about signal energy, scale, and distribution
Fifteen RR features	represent local rate, premature beats, rhythm irregularity, and recent interval trend using information reproducible during streaming
Normal-only autoencoder	supplies a learned measure of disagreement with normal ECG structure without requiring arrhythmia labels
Two-stage hierarchy	allows unsupported abnormal beats to count in binary testing without forcing them into one of the five supported subtype labels
Extra Trees models	handle mixed feature scales and non-linear interactions with reproducible tabular-model training
Temporal split with gap	evaluates later signal from each record while removing direct boundary overlap between development and test beats

The architecture is therefore best described as a known-patient temporal continuation system. Its final performance numbers describe later MLII signal from patients whose earlier MLII signal was available during model development. They should not be presented as unseen-patient generalisation without a separate patient-disjoint evaluation.

6 Experimental Method

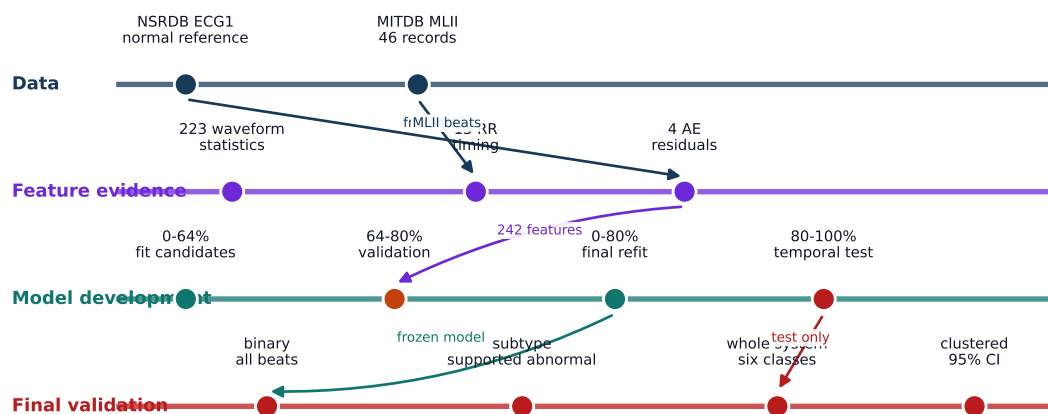
The final experiment measures one specific use case: the system has already seen earlier MLII ECG from a patient, and it must classify later ECG from that same patient. This is a known-patient temporal holdout. It is not a new-patient external validation and it is not a shuffled random beat split.

6.1 Method overview

Figure 6.1 gives the complete final method. The normal autoencoder was already frozen from NSRDB. The supervised experiment then used only MITDB MLII beats. Candidate models were selected with earlier temporal regions, refitted before the test boundary, and evaluated only on the final 20% timeline.

Experimental method for the final temporal-holdout run

One-lead MLII experiment: select with earlier beats, freeze decisions, score only the final 20% timeline.



The test timeline is not used for candidate choice, threshold choice, or final fitting in this rerun.

Figure 6.1: Experimental workflow for the final run. The method keeps source data, feature construction, temporal splitting, model selection, final fitting, temporal scoring, and subject-clustered validation as separate steps.

6.2 Research objective and success conditions

The final experiment asks:

After the model has access to the earlier ECG of each patient, how accurately does

it classify that patient's later ECG?

The primary outcome is whole-system six-class accuracy on the final 20% temporal holdout. The planned success conditions were:

1. pooled whole-system accuracy at least 90%;
2. lower bound of the equal-subject 95% confidence interval above 90%; and
3. total confidence-interval width no greater than three percentage points.

Per-class recall and macro F1 were also checked so that a high overall score could not hide failure on a smaller arrhythmia class.

6.3 Input data and feature construction

The method uses one ECG channel at every stage, but the two data sources have different roles. NSRDB supplies healthy windows for the frozen normal autoencoder. MITDB supplies MLII beat windows, binary labels, six-class labels, and the final temporal test. No NSRDB beat is scored in the final test.

Table 6.1: Exact evidence supplied to the final supervised models.

Evidence block	Count	Role
MLII morphology and spectrum	223 values	Describes the beat-centred 512-sample MLII waveform. This is the direct ECG evidence for both supervised stages.
RR timing context	15 values	Describes previous RR, next RR, local timing, and recent rhythm history. In live use the beat is scored only after this next-RR context is available; labels are never used during prediction.
Frozen autoencoder residuals	4 values	Summarises how the frozen NSRDB-trained normal autoencoder reconstructs the same MLII beat.
Total per MITDB beat	242 values	The binary detector and subtype classifier receive the same one-lead feature vector.

This feature construction matches the saved final model bundle: `lead_count` is one, `required_lead` is MLII, `rr_feature_count` is 15, and `ae_feature_count` is four. The method therefore does not use a second MITDB lead during training, validation, temporal testing, or deployment.

6.4 Temporal split

The eligible supervised records are the 46 MITDB records that contain MLII. Records 102 and 104 were excluded before fitting because their available leads are V5 and V2, not MLII. Records 201 and 202 are treated as one subject cluster for uncertainty analysis because they belong to the same person.

Every eligible record is kept in chronological order:

1. the first 64% is the candidate-training region;
2. a 16-beat gap is removed before validation during candidate selection;
3. the 64–80% region is the validation region;
4. a 16-beat gap is removed before the final temporal test; and
5. the final 20% is the fixed temporal test.

For final fitting, the selected binary and subtype models are refitted on the earlier 80% development region after the 16-beat boundary gap before the test is removed. The test segment itself remains intact.

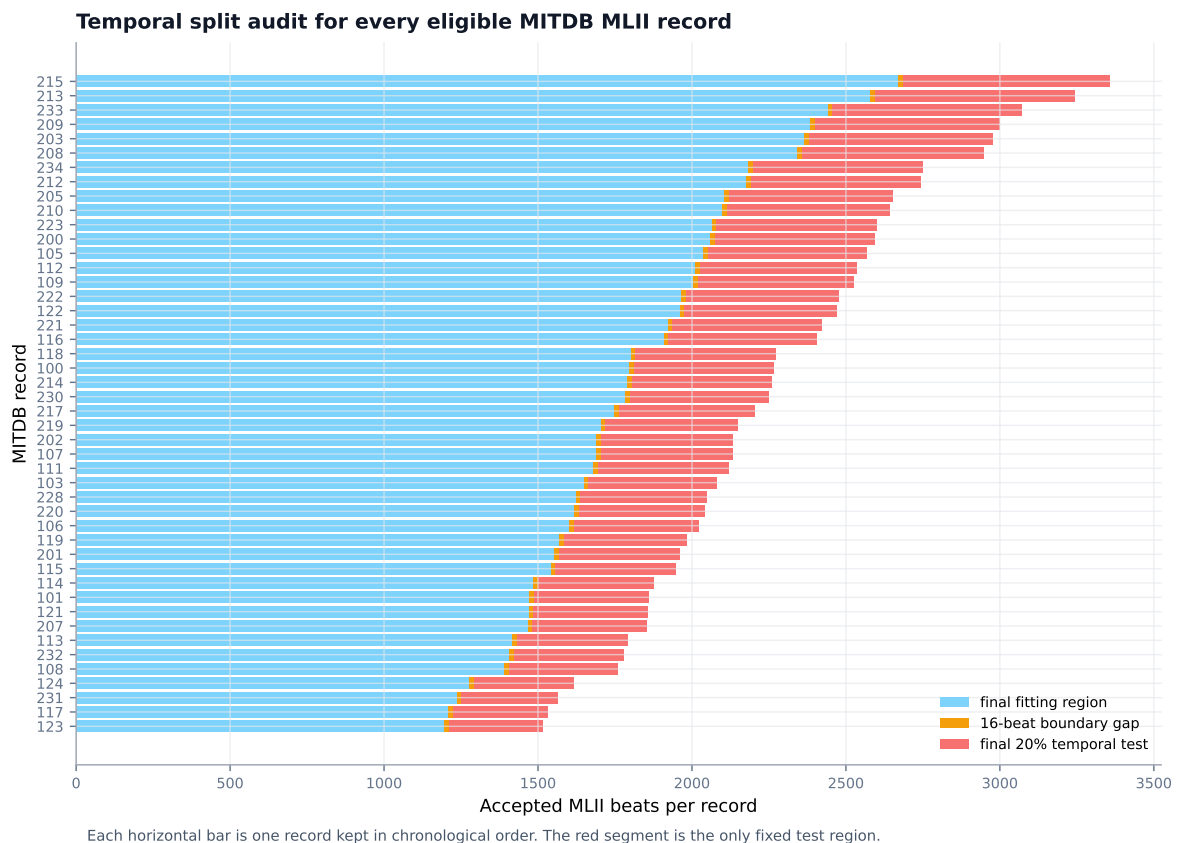


Figure 6.2: Temporal split audit for every eligible MITDB MLII record. Blue beats are available for final fitting, yellow beats are the boundary gap before the test, and red beats are the held-out final 20% temporal test.

The final held-out test contains 20,988 accepted beats. These are later beats from known patients, not beats from unseen patients. The method prevents signal overlap across the test boundary, but it intentionally keeps patient overlap.

6.5 Model selection and final fitting

Model selection uses only the earlier temporal validation region. In this rerun, the final 20% test region does not choose the candidate, threshold, or hyperparameters.

Three binary Extra Trees candidates were fitted on the earlier candidate region and ranked by validation balanced accuracy, then by F1. Three subtype Extra Trees candidates were fitted only on supported abnormal candidate beats and ranked by validation macro F1, then by accuracy. Figure 6.3 shows the validation scores that selected the final pair.

Validation-region model selection

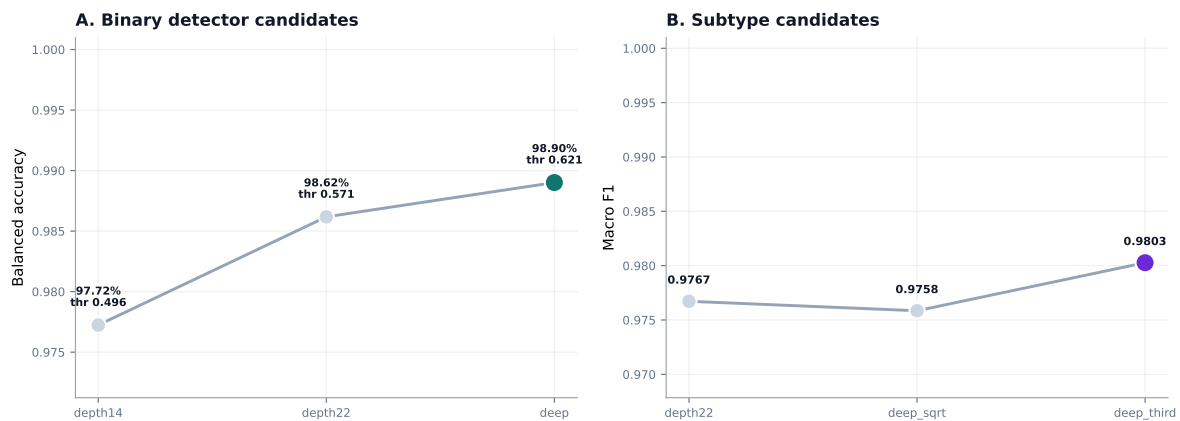


Figure 6.3: Model selection on the 64–80% validation region. The binary detector is selected by balanced accuracy and F1; the subtype classifier is selected by macro F1 and accuracy.

The selected binary detector was the fully grown 240-tree candidate with square-root feature sampling, minimum leaf size one, and record-weighting power 0.35. Its validation-selected abnormal threshold was 0.6208333. The selected subtype classifier was the fully grown 260-tree candidate with one-third feature sampling, minimum leaf size one, and record-weighting power 0.25. After selection, the binary model was refitted on all development beats, whereas the subtype model was refitted only on supported abnormal development beats.

6.6 Evaluation order

The deployed decision is hierarchical. First, the binary detector estimates whether a beat is abnormal. If the abnormal probability is not above the fixed threshold, the final output is N. If the beat passes the gate, the subtype classifier chooses PVC, PAC, LBBB, RBBB, or AFib.

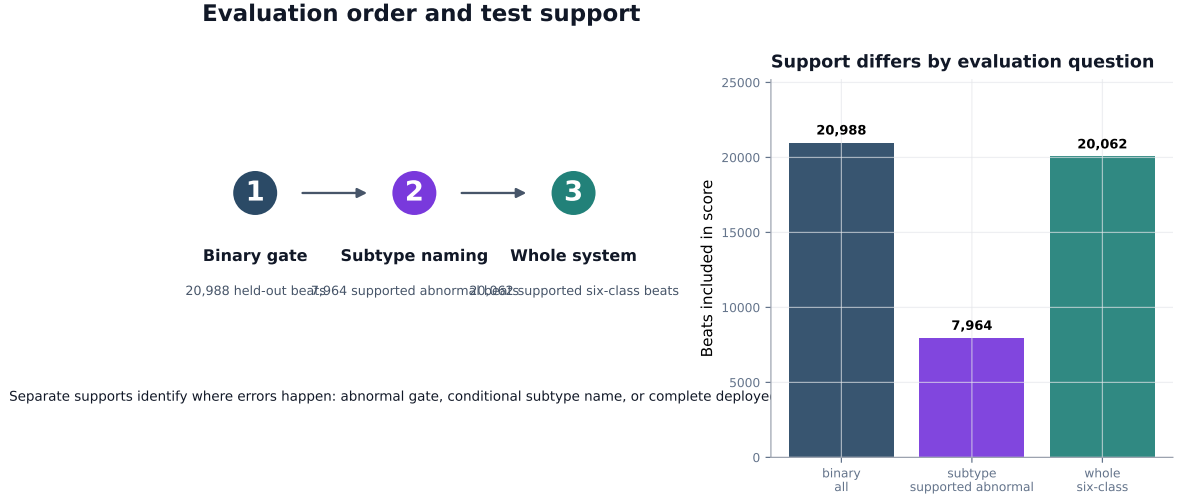


Figure 6.4: Evaluation order used by the final method. Binary, conditional subtype, and whole-system evaluations answer different questions and therefore use different supports.

The temporal evaluation reports three supports:

1. binary abnormality detection on all 20,988 held-out beats, including unsupported abnormal MITDB symbols;
2. conditional subtype classification on 7,964 true supported abnormal beats; and
3. whole-system N/PVC/PAC/LBBB/RBBB/AFib classification on 20,062 supported held-out beats.

The 926 unsupported abnormal beats are useful for testing the binary normal/abnormal gate, but they are not scored in the six-class whole-system confusion matrix.

6.7 Metrics

For a binary outcome,

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}, \quad (6.1)$$

$$\text{Sensitivity} = \frac{TP}{TP + FN}, \quad (6.2)$$

$$\text{Specificity} = \frac{TN}{TN + FP}, \quad (6.3)$$

$$\text{Balanced accuracy} = \frac{\text{Sensitivity} + \text{Specificity}}{2}. \quad (6.4)$$

For class c ,

$$\text{Precision}_c = \frac{TP_c}{TP_c + FP_c}, \quad (6.5)$$

$$\text{Recall}_c = \frac{TP_c}{TP_c + FN_c}, \quad (6.6)$$

$$F1_c = \frac{2 \text{Precision}_c \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c}. \quad (6.7)$$

Macro F1 is the unweighted mean of class F1 values. It gives PAC the same weight as N even though N contains many more beats.

Two accuracy summaries are reported. Pooled beat accuracy gives every beat equal weight. Subject-macro accuracy first calculates accuracy inside each of the 45 eligible subject clusters and then gives every subject equal weight:

$$A_{\text{macro}} = \frac{1}{45} \sum_{s=1}^{45} A_s. \quad (6.8)$$

The subject-macro value is primary for uncertainty because a long record should not behave like many independent patients.

6.8 Subject-clustered confidence interval

The primary 95% confidence interval uses a non-parametric percentile bootstrap:

1. keep every beat from one subject together;
2. treat records 201 and 202 as one subject cluster;
3. sample 45 subject clusters with replacement;
4. calculate the equal-subject mean accuracy;
5. repeat 10,000 times using seed 20260801; and
6. take the 2.5th and 97.5th percentiles.

Resampling individual beats would create an artificially narrow interval because adjacent beats from the same recording are correlated. The clustered bootstrap keeps that dependence [23].

6.9 Tests against the 90% target

The confidence interval is the primary statistical evidence. Two non-parametric tests provide sensitivity analyses. The exact one-sided sign test asks whether more than half of subjects exceed 90%. The one-sided Wilcoxon signed-rank test asks whether subject accuracies are shifted above 90%. These tests are interpreted together with the confidence interval, interval width, per-class recall, macro F1, and subject accuracy distribution.

6.10 Reproducibility and leakage controls

The final experiment saves the temporal boundary audit for every record, the candidate scores, selected models, probability threshold, final confusion matrix, per-class counts, per-subject accuracies, bootstrap seed, and headline metrics. The rerun used seed 20260801 and did not use the final 20% segment for candidate selection, threshold selection, or final refitting.

The final 20% segment had been observed during earlier system development, so the evaluation is not a pristine first-use external test. Patient overlap is also intentional: the test is later ECG from known patients. Therefore, the method supports a patient-adapted temporal-performance claim, not a patient-independent generalisation claim.

7 Results and Statistical Validation

This chapter reports the fixed temporal holdout experiment saved by the final experiment artifact. The reported values match that artifact. The evaluation is not a new-patient external validation. It is a patient-adapted temporal test: earlier MLII ECG from each eligible MITDB record is available during training and selection, and only the final 20% timeline is scored as unseen signal from the same known patients.

7.1 What was tested

Figure 7.1 summarises the actual final test. The test data came only from the MIT-BIH Arrhythmia Database. Records 102 and 104 were excluded because they do not contain MLII. The remaining 46 records with MLII represent 45 subject clusters, with records 201 and 202 treated as one subject. No NSRDB beat is part of the final temporal test. NSRDB contributes only the already frozen normal autoencoder, which creates four residual features for each MITDB MLII beat.

How the final system was tested

Fixed within-patient temporal holdout: earlier MLII from each patient trains/selects the model; only the final 20% timeline is scored.

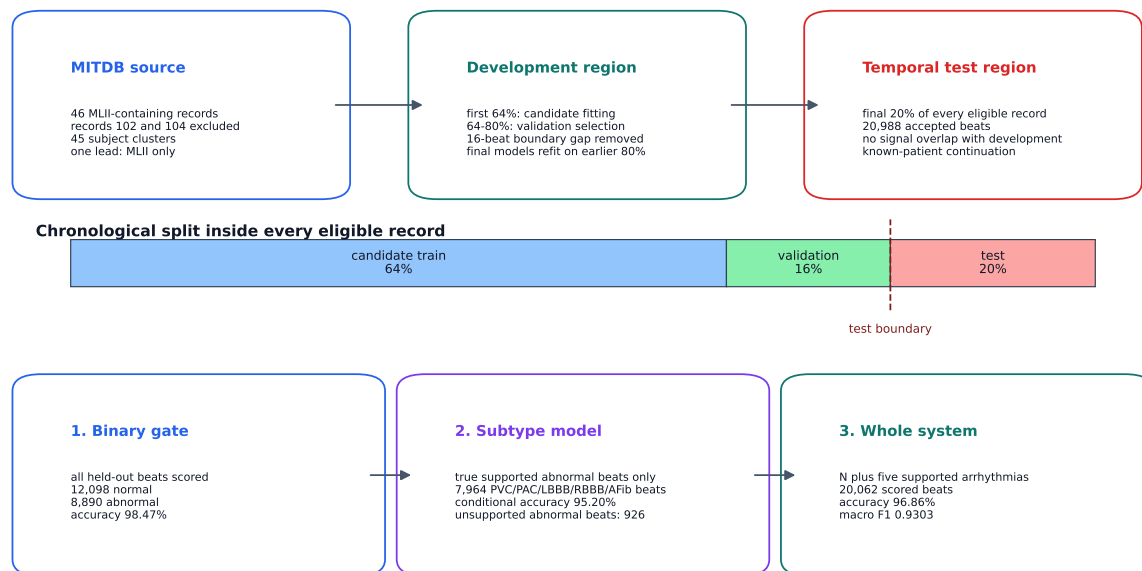


Figure 7.1: Final testing protocol. Each eligible MITDB MLII record is split in time. Candidate fitting and validation use only earlier beats; the final 20% timeline is held out for scoring.

Table 7.1: Data used by each final evaluation.

Evaluation	Test support	Meaning
Binary abnormality detection	20,988 held-out beats: 12,098 normal and 8,890 abnormal	Tests whether the gate separates normal from abnormal. Unsupported abnormal MITDB symbols are included here because they are still abnormal beats.
Conditional subtype classification	7,964 supported abnormal beats	Tests the subtype model only on true PVC, PAC, LBBB, RBBB, and AFib beats. This ignores the binary gate so subtype naming can be inspected separately.
Whole-system six-class output	20,062 supported beats	Tests the deployed hierarchy after the binary gate: N/PVC/PAC/LBBB/RBBB/AFib. The 926 unsupported abnormal beats are not included in the six-class confusion matrix.

What data was used for the final temporal test

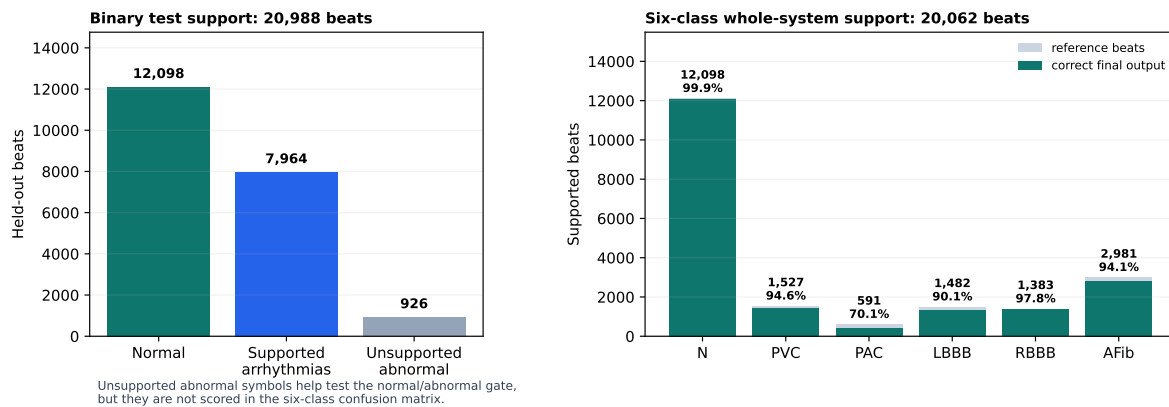


Figure 7.2: Composition of the held-out temporal test data. The binary test uses all accepted held-out beats. The six-class whole-system test uses normal beats and the five supported abnormal classes.

7.2 Model selection before final testing

In this rerun, the final 20% segment did not choose the model or the threshold. Candidate selection used the 64–80% validation region. The selected binary model was a fully grown 240-tree Extra Trees ensemble using square-root feature sampling, leaf size one, and

record-weighting power 0.35. Its selected abnormal threshold was 0.6208333, and its validation balanced accuracy was 98.90%. The selected subtype model was a fully grown 260-tree ensemble using one-third feature sampling, leaf size one, and record-weighting power 0.25. Its validation macro F1 was 0.9803. After selection, both supervised models were refitted on the earlier 80% development region, excluding the 16-beat boundary gap before the temporal test.

7.3 Headline temporal performance

Table 7.2: Final temporal results. Confidence intervals use equal-subject scores, not pooled beats.

Outcome	Pooled score	Subject-macro	95% CI	Width
Abnormality accuracy	98.47%	98.53%	97.16–99.41%	2.26 pp
Conditional subtype accuracy	95.20%	93.80%	87.71–98.58%	10.86 pp
Whole-system accuracy	96.86%	96.90%	94.02–99.05%	5.03 pp

The binary gate reached 98.47% pooled accuracy, 98.22% balanced accuracy, 96.59% sensitivity, 99.85% specificity, 99.79% precision, and binary F1 of 0.9817. Conditional subtype accuracy on true supported abnormal beats was 95.20%, with macro F1 of 0.9325. After the gate and subtype model were combined, whole-system six-class accuracy was 96.86% and macro F1 was 0.9303.

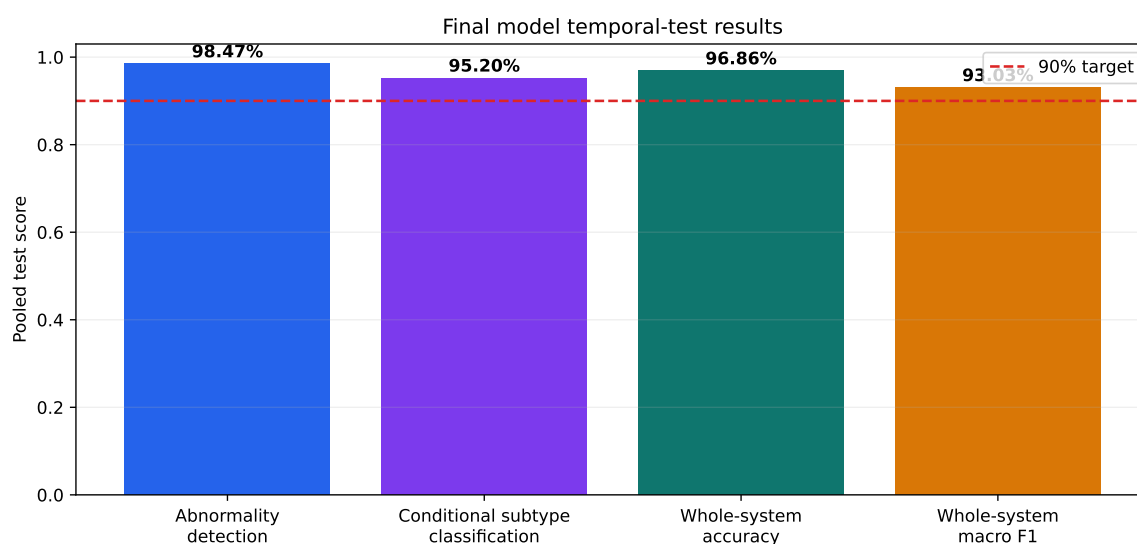


Figure 7.3: Pooled temporal-test results. Binary accuracy, conditional subtype accuracy, whole-system accuracy, and whole-system macro F1 are all above 90%.

The primary point estimate exceeded the 90% target, and the lower bound of the equal-subject 95% confidence interval was also above 90%. The complete planned statistical goal

was not fully met because the whole-system confidence interval was 5.03 percentage points wide. The planned maximum was three percentage points.

7.4 Subject-clustered validation

The statistical validation treats subjects, not beats, as the independent units. Each bootstrap replicate sampled complete subject clusters with replacement, kept records 201 and 202 inside one subject cluster, and then calculated equal-subject accuracy. This prevents the 20,988 related beats from making the confidence interval falsely narrow.

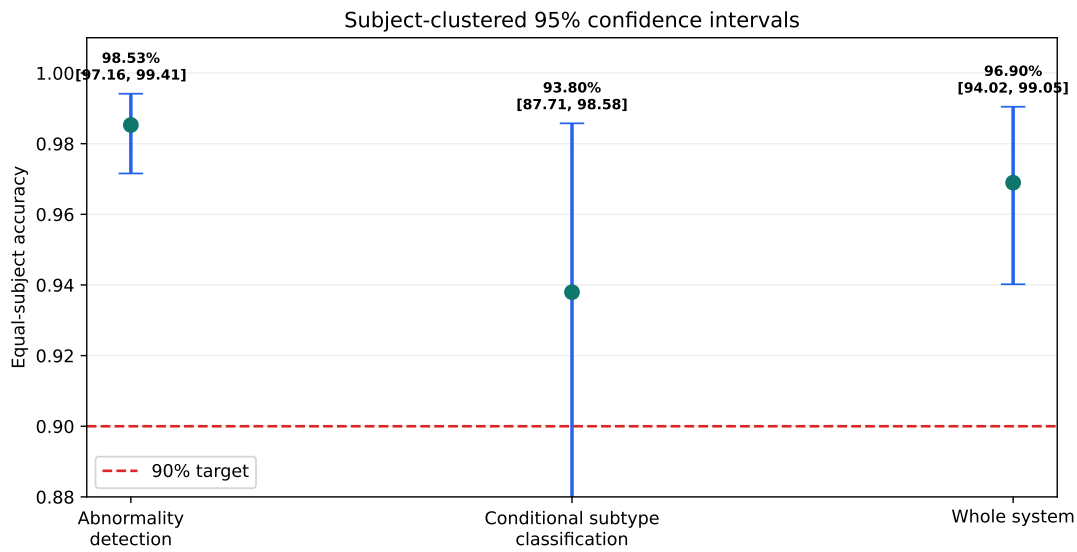


Figure 7.4: Equal-subject estimates and 95% subject-cluster bootstrap intervals. The whole-system lower bound remains above 90%, but the interval width is larger than planned.

Forty-two of 45 subject clusters had whole-system accuracy above 90%. The exact one-sided sign test gave $p = 4.33 \times 10^{-10}$ for the hypothesis that more than half of subjects exceed 90%. A secondary one-sided Wilcoxon signed-rank test gave statistic 903 and $p = 5.63 \times 10^{-6}$ for a positive shift above 90%. These tests support the conclusion that performance is above 90% for this known-patient temporal problem, but the confidence interval remains the primary effect-size evidence.

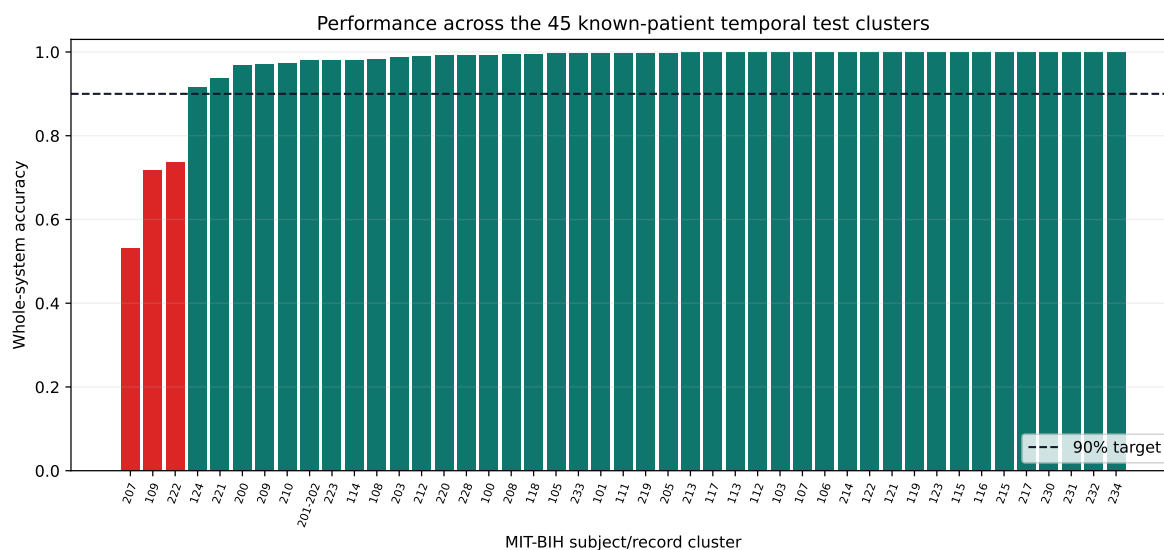


Figure 7.5: Whole-system accuracy for all 45 subject clusters. Red bars are below 90%; 42 clusters are above the target.

The below-target clusters were 207 (53.10%), 109 (71.74%), and 222 (73.59%). Cluster 207 mainly reflects subtype confusion, while cluster 222 has a weaker binary result. The next-lowest cluster, 124, reached 91.64%. This subject variation explains why the equal-subject confidence interval is much wider than a pooled beat-level summary would suggest.

7.5 Stage-by-stage results

The binary gate was tested on all 20,988 held-out beats. It correctly detected 8,587 of 8,890 abnormal beats and correctly rejected 12,080 of 12,098 normal beats. The remaining binary errors were 303 missed abnormal beats and 18 false alarms. This stage meets the aggregate accuracy, lower-bound, and interval width targets.

The subtype model was evaluated independently on 7,964 true supported abnormal beats. It correctly named 95.20% of them before the binary gate was applied. This explains why the conditional subtype score is higher than the final end-to-end subtype recall for some arrhythmias. A beat can be named correctly by the subtype model but still be sent to normal by the binary gate.

7.6 Results for each arrhythmia

Table 7.3: End-to-end class performance on real labeled temporal test beats.

Type	True	Correct	Precision	Recall	F1	90% recall
PVC	1,527	1,445	85.30%	94.63%	0.8972	Reached
PAC	591	414	98.57%	70.05%	0.8190	Not reached
LBBB	1,482	1,336	95.09%	90.15%	0.9255	Reached
RBBB	1,383	1,352	99.78%	97.76%	0.9876	Reached
AFib	2,981	2,806	98.77%	94.13%	0.9639	Reached

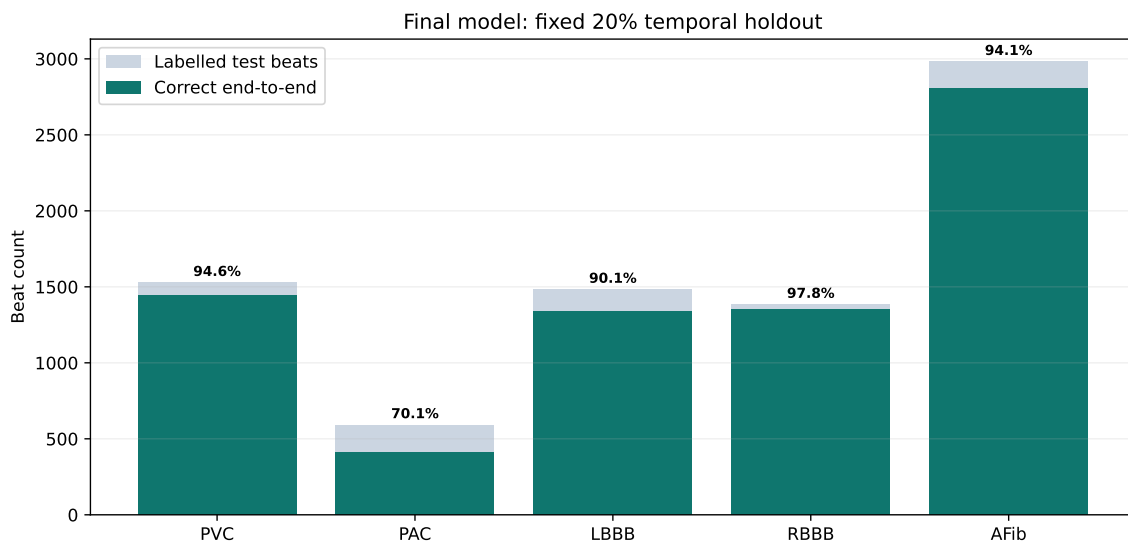


Figure 7.6: Reference and correctly classified arrhythmia counts. PAC is the only supported abnormal class with end-to-end recall far below 90%.

PVC, LBBB, RBBB, and AFib exceeded 90% end-to-end recall. PAC did not. Of 591 PAC beats, the binary detector marked 540 abnormal, the independent subtype model named 445 correctly before gating, and only 414 survived both decisions with the correct final output. Therefore, PAC loss comes from both the binary gate and the subtype model, with subtype confusion contributing more. The aggregate system accuracy must not be interpreted as equal strength for every arrhythmia.

7.7 Whole-system confusion matrix

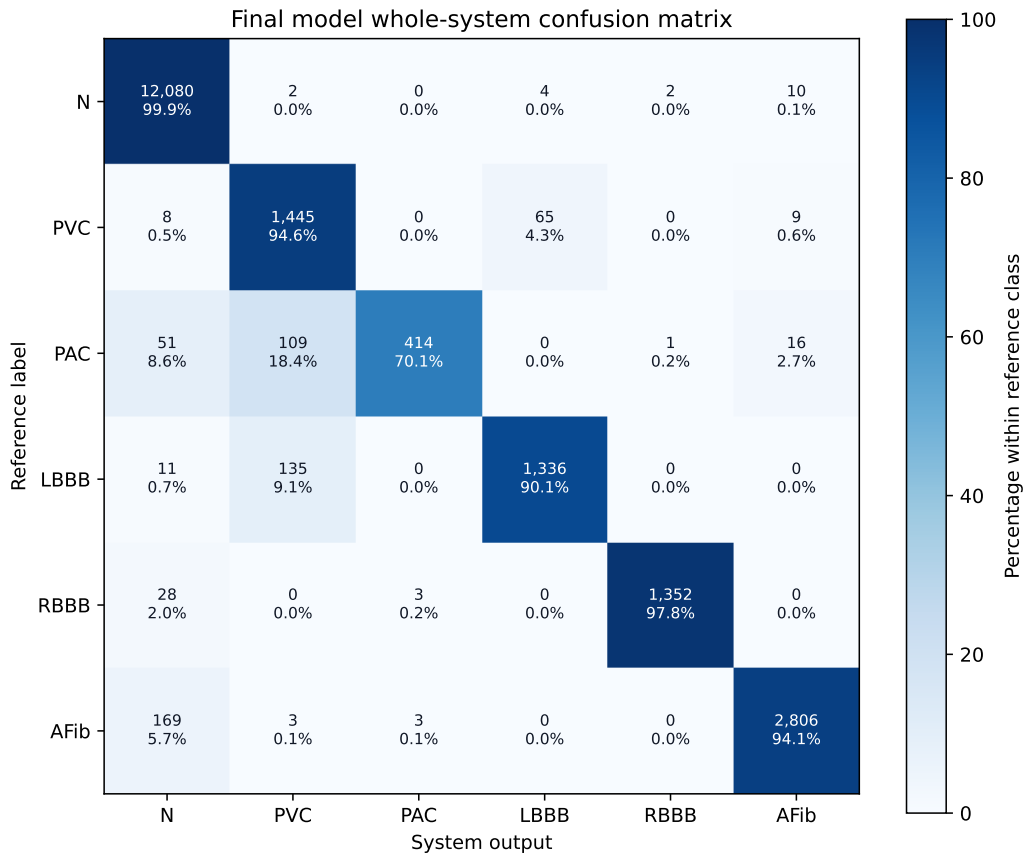


Figure 7.7: Confusion matrix for 20,062 supported temporal-test beats. Counts and row percentages are shown together.

The diagonal dominates the confusion matrix, but the remaining errors show the main failure modes. The largest subtype confusions are 135 LBBB beats assigned to PVC, 109 PAC beats assigned to PVC, and 65 PVC beats assigned to LBBB. The binary gate sends 169 AFib beats and 51 PAC beats to normal. These errors are clinically important even though the aggregate whole-system accuracy is high.

7.8 Interpretation of the temporal result

The final result should be read as evidence for temporal continuation after patient exposure. Earlier ECG from the same patient provides morphology, lead orientation, and rhythm context that can make later beats easier to classify. The chronological boundary and 16-beat gap prevent direct signal overlap, but they do not remove patient identity. Therefore, the 96.86% whole-system accuracy is valid for known-patient temporal holdout performance. It is not proof of cold-start performance on completely unseen patients.

7.9 External INCART unseen-dataset check

After the final MITDB temporal-holdout system had been fixed, a separate external check was run on the St Petersburg INCART 12-lead Arrhythmia Database. This is not the primary thesis benchmark and it is not used for model selection. It is included to show how the frozen system behaves on an unseen source database with a different acquisition setting. Lead II is used because it is the closest practical counterpart to MITDB MLII, but it is not identical to MLII. Therefore, this section measures both external-patient shift and lead/device-domain shift.

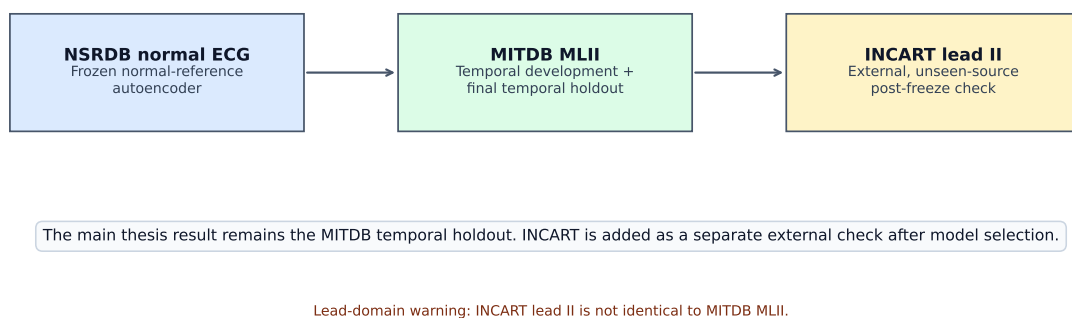


Figure 7.8: Separation between the main MITDB temporal result and the INCART external check. The external result is reported after model selection and is not used to tune the frozen system.

Table 7.4: Limited external INCART check on the latest saved evaluation artifact.

Item	Value
Dataset	St Petersburg INCART 12-lead Arrhythmia Database
Lead and mode	II, offline
Records scored	1 (I01)
Annotated beats	120
Saved artifact time	2026-08-24T09:45:26.627767+00:00
Decision threshold	0.6208
Accuracy	98.33%
Balanced accuracy	99.10%
Sensitivity / specificity	100.00% / 98.20%
Precision / F1	81.82% / 90.00%
AUROC / AUPRC	1.0000 / 1.0000
Confusion counts	TP=9, TN=109, FP=2, FN=0
Subtype macro F1	0.9691

On this limited INCART subset, the frozen hierarchy reached 98.33% binary accuracy and 99.10% balanced accuracy. All 9 annotated abnormal beats in the subset were detected and

109 normal beats were rejected, while 2 normal beats were false alarms and 0 abnormal beats were missed. Because the subset is small and contains only 120 accepted beats from 1 INCART record, this result should be read as an external sanity check, not as a definitive external clinical validation.

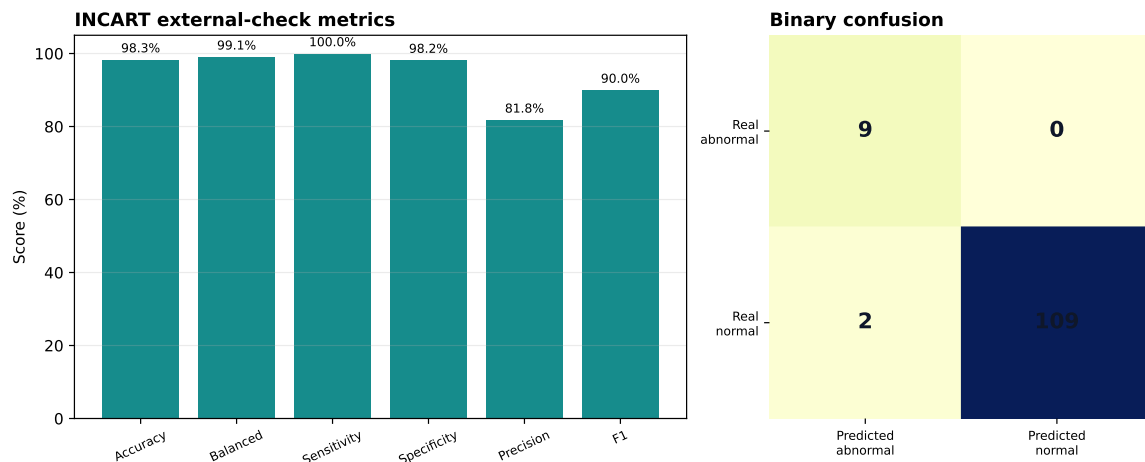


Figure 7.9: Metric bars and binary confusion matrix for the limited INCART external check. Balanced accuracy is shown beside pooled accuracy because the subset is class-imbalanced.

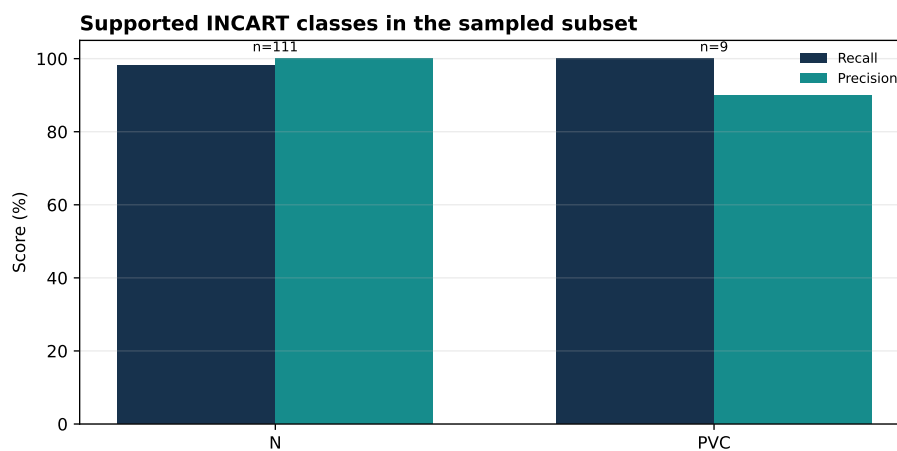
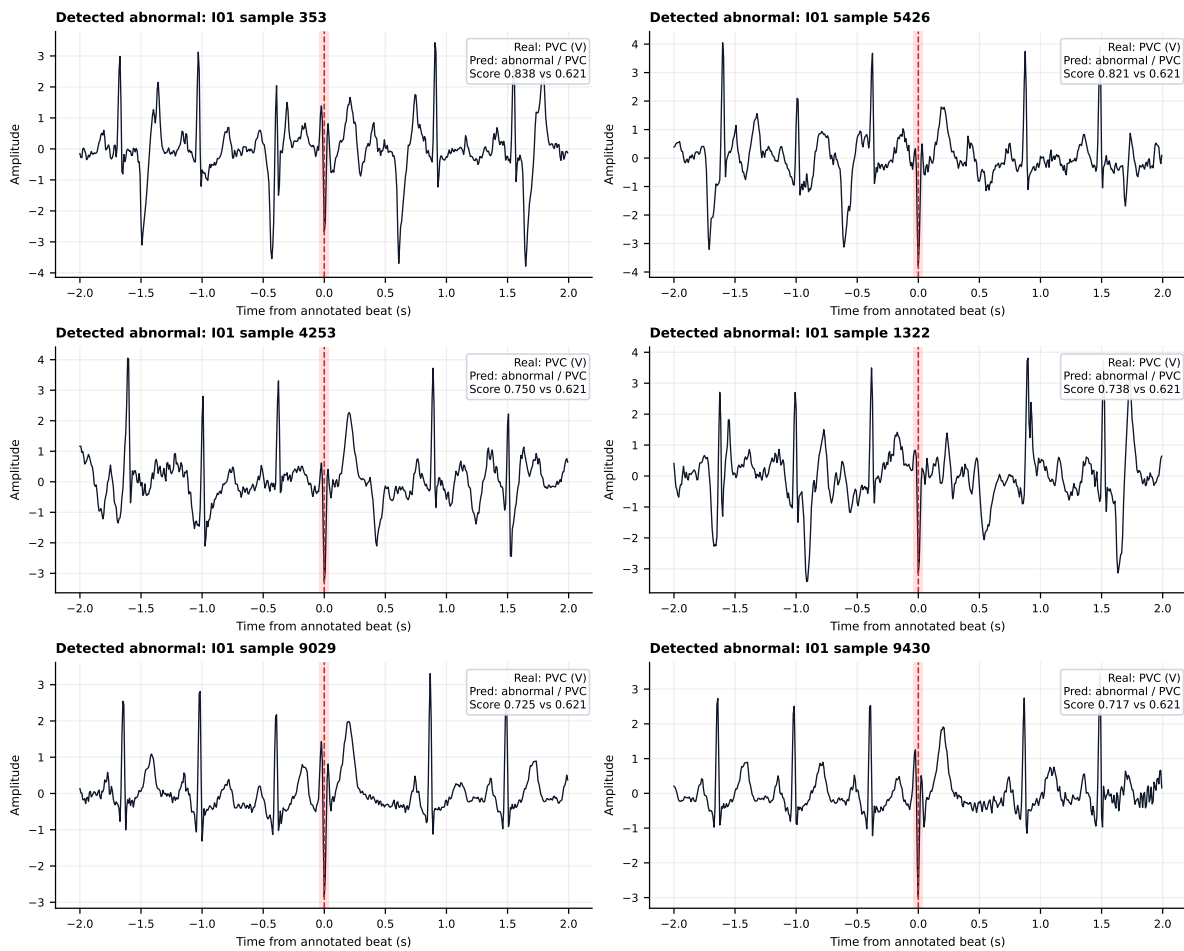


Figure 7.10: Class-level precision and recall for classes that appear in the limited INCART subset. Missing classes are not plotted because they have zero support in this saved evaluation.

Table 7.5: Example annotated INCART beats used for visual inspection.

No.	Record/sample	Real type	Predicted decision/type	Score
1	I01/353	PVC (V)	Abnormal / PVC	0.838
2	I01/5426	PVC (V)	Abnormal / PVC	0.821
3	I01/4253	PVC (V)	Abnormal / PVC	0.750
4	I01/1322	PVC (V)	Abnormal / PVC	0.738
5	I01/9029	PVC (V)	Abnormal / PVC	0.725
6	I01/9430	PVC (V)	Abnormal / PVC	0.717

**Figure 7.11:** Example INCART ECG segments centred on annotated beats. Each panel shows the real annotation type, the model's abnormal/normal decision, the predicted anomaly type, and the score-threshold comparison.

7.10 Answers to the research questions

RQ1 The frozen normal-only NSRDB autoencoder supplies four reconstruction-error features to both MLII supervised ensembles. A held-out ablation was not performed, so its

independent contribution is not quantified.

- RQ2** Abnormality detection reached 98.47%, conditional subtype accuracy reached 95.20%, and whole-system accuracy reached 96.86% on the fixed temporal test.
- RQ3** The whole-system equal-subject estimate was 96.90%, with a 94.02–99.05% subject-clustered 95% interval. Its lower bound is above 90%.
- RQ4** The complete goal was not reached. The point estimate and lower confidence bound passed, but the interval width was 5.03 percentage points instead of at most three and PAC recall was 70.05%. A limited post-freeze INCART external check was added in Section 7.9, but it is too small to replace a full external clinical validation.

8 Discussion and Conclusion

8.1 Core finding

The final one-lead system reached high overall accuracy on later MLII ECG from patients whose earlier MLII ECG was used for training. Whole-system pooled accuracy was 96.86%, equal-subject accuracy was 96.90%, and the subject-clustered 95% confidence interval was 94.02–99.05%. Its lower bound is above 90%, and two non-parametric subject-level tests also reject a 90% null hypothesis.

This is a genuine chronological holdout, not a shuffled-beat calculation. The final 20% was excluded from fitting and selection, and a 16-beat gap reduced contact between neighbouring windows. However, it is a patient-adapted result: the same patients appear in the first 80% training region. It must not be described as 96.86% accuracy on unseen patients.

A limited post-freeze INCART check was added after the main final experiment. It used lead II from an unseen source database and is useful for demonstrating external-domain behaviour, including example ECG segments. It does not replace a large patient-level external validation study.

8.2 Were the goals reached?

- **Aggregate accuracy above 90%: reached.** Abnormality, conditional subtype, whole-system accuracy, and whole-system macro F1 all exceed 90%.
- **Whole-system 95% interval above 90%: reached.** The lower bound is 94.02%.
- **Interval width at most three points: not reached.** The width is 5.03 points.
- **Every arrhythmia above 90% recall: not reached.** PAC recall is 70.05%.
- **More than 90% accuracy on completely unseen patients: partially checked, not fully validated.** The limited INCART check exceeded 90% on its saved subset, but it is too small to establish patient-level external generalisation.

8.3 Why the final architecture worked

The system works as a one-channel system. The autoencoder uses only the first stored channel of NSRDB, while every supervised and deployed input is MLII. The NSRDB header calls its channel ECG1 but does not document an anatomical Lead-II placement. For that reason, this thesis does not claim that the two source channels have the same electrical axis.

Removing MITDB records without MLII also prevents the supervised system from silently changing viewpoints.

The model combines evidence that explains different parts of the ECG. Beat bins and derivatives describe shape. Spectral features describe frequency content. Causal RR history describes timing. Normal-only reconstruction describes mismatch from healthy structure. Extra Trees learn non-linear links between these features without assuming a straight-line decision boundary. Earlier labelled ECG also gives a patient-specific reference, which makes persistent bundle-branch and rhythm patterns easier to recognise later in the same recording.

8.4 Remaining weaknesses

PAC remains difficult because it can look like normal rhythm or PVC and often depends on subtle timing evidence. The final system correctly classified 414 of 591 PAC beats end to end. Its PAC precision is high, but sensitivity is low. Improvement requires more independent PAC patients and a larger external test, not more tuning against the temporal test or the limited INCART check already reported here.

The interval width is controlled by 45 independent subject clusters, including three below-target clusters. Adding more beats from the same records would produce a misleadingly narrow beat-level interval without adding independent evidence. A genuinely shorter interval requires more independent patients with enough examples from all classes.

8.5 Limitations

1. The temporal study has patient overlap by design and cannot establish cold-start patient generalisation.
2. The supervised data represent 45 MITDB subject clusters; no external hospital cohort was used. The added INCART check is an external database check, not a hospital-cohort validation study.
3. The healthy autoencoder used complete recordings from all 18 NSRDB subjects. Its first stored channel is named ECG1 in the WFDB headers, but the electrode placement is not documented as Lead II.
4. The independent benefit of the four autoencoder features was not tested with a final held-out ablation.
5. PAC end-to-end recall is 70.05%.
6. Unsupported abnormal symbols remain in the binary result but are excluded from the supported six-class result.
7. Records 102 and 104 were excluded because they contain V5 and V2, not MLII. Record 114 uses its stored MLII channel by signal name rather than assuming a fixed channel number.

8. Offline evaluation uses annotated R peaks. Live R-peak detection can add errors that are not included in the offline temporal result.
9. Evaluation is beat based, not a clinical episode or patient-level diagnosis study.

These limits follow the principle that development performance and clinical evidence must be reported separately [24].

8.6 Required next study

A future study should add many independent patients, reserve an external patient-level database for one final test, pre-register a lower 95% bound above 90%, keep the three-point width target, and set minimum recall for each arrhythmia. All thresholds must be selected without the external test. The study should also compare the full model with and without autoencoder residuals, and validate real-time R-peak detection and Lead-II hardware placement.

8.7 Final statement

The thesis demonstrates a one-lead, patient-adapted temporal ECG system with 96.86% whole-system accuracy and a 94.02–99.05% subject-clustered 95% interval. This is meaningful evidence for later MLII ECG from known patients. The added INCART result is a useful external-source check, but it is not evidence of equivalent performance for all new patients, and it does not solve PAC sensitivity. The aggregate final-study goal is reached with clearly stated statistical and clinical limits. The system remains for research use only.

Bibliography

- [1] P. Kligfield, L. S. Gettes, J. J. Bailey, R. Childers, B. J. Deal, E. W. Hancock, G. van Herpen, J. A. Kors, P. Macfarlane, D. M. Mirvis, O. Pahlm, P. Rautaharju, and G. S. Wagner, “Recommendations for the standardization and interpretation of the electrocardiogram: Part i,” *Journal of the American College of Cardiology*, vol. 49, no. 10, pp. 1109–1127, 2007.
→ [p1], [p22]
- [2] G. B. Moody and R. G. Mark, “The impact of the mit–bih arrhythmia database,” *IEEE Engineering in Medicine and Biology Magazine*, vol. 20, no. 3, pp. 45–50, 2001.
→ [p1], [p6], [p11]
- [3] P. de Chazal, M. O’Dwyer, and R. B. Reilly, “Automatic classification of heartbeats using ecg morphology and heartbeat interval features,” *IEEE Transactions on Biomedical Engineering*, vol. 51, no. 7, pp. 1196–1206, 2004.
→ [p1], [p6], [p9], [p20]
- [4] A. L. Goldberger, L. A. N. Amaral, L. Glass, J. M. Hausdorff, P. C. Ivanov, R. G. Mark, J. E. Mietus, G. B. Moody, C.-K. Peng, and H. E. Stanley, “Physiobank, physiotoolkit, and physionet: Components of a new research resource for complex physiologic signals,” *Circulation*, vol. 101, no. 23, pp. e215–e220, 2000.
→ [p6], [p11]
- [5] G. B. Moody and R. G. Mark, “Mit–bih arrhythmia database.” <https://physionet.org/content/mitdb/1.0.0/>, 2005.
→ [p6], [p11], [p12]
- [6] G. B. Moody, “Mit–bih normal sinus rhythm database.” <https://physionet.org/content/nsrdb/1.0.0/>, 1999.
→ [p6], [p9], [p11], [p31]
- [7] S. Mousavi and F. Afghah, “Inter- and intra-patient ecg heartbeat classification for arrhythmia detection: A sequence to sequence deep learning approach,” in *IEEE International Conference on Acoustics, Speech and Signal Processing*, pp. 1308–1312, 2019.
→ [p6], [p7], [p9], [p20]
- [8] J. Zhang, A. Liu, D. Liang, X. Chen, and M. Gao, “Interpatient ecg heartbeat classification with an adversarial convolutional neural network,” *Journal of Healthcare Engineering*, vol. 2021, p. 9946596, 2021.
→ [p6], [p7], [p9], [p25]
- [9] H. Huang, J. Liu, Q. Zhu, R. Wang, and G. Hu, “Detection of inter-patient left and right bundle branch block heartbeats in ecg using ensemble classifiers,” *BioMedical Engineering OnLine*, vol. 13, p. 72, 2014.
→ [p7], [p9]

- [10] S. Kiranyaz, T. Ince, and M. Gabbouj, “Real-time patient-specific ecg classification by 1-d convolutional neural networks,” *IEEE Transactions on Biomedical Engineering*, vol. 63, no. 3, pp. 664–675, 2016. → [p7]
- [11] U. R. Acharya, H. Fujita, O. S. Lih, Y. Hagiwara, J. H. Tan, and M. Adam, “Automated detection of arrhythmias using different intervals of tachycardia ecg segments with convolutional neural network,” *Information Sciences*, vol. 405, pp. 81–90, 2017. → [p7]
- [12] A. Y. Hannun, P. Rajpurkar, M. Haghpanahi, G. H. Tison, C. Bourn, M. P. Turakhia, and A. Y. Ng, “Cardiologist-level arrhythmia detection and classification in ambulatory electrocardiograms using a deep neural network,” *Nature Medicine*, vol. 25, pp. 65–69, 2019. → [p7]
- [13] A. H. Ribeiro, M. H. Ribeiro, G. M. M. Paixao, D. M. Oliveira, P. R. Gomes, J. A. Canazart, M. P. S. Ferreira, C. R. Andersson, P. W. Macfarlane, W. Meira, T. B. Schön, and A. L. P. Ribeiro, “Automatic diagnosis of the 12-lead ecg using a deep neural network,” *Nature Communications*, vol. 11, p. 1760, 2020. → [p7]
- [14] G. E. Hinton and R. R. Salakhutdinov, “Reducing the dimensionality of data with neural networks,” *Science*, vol. 313, no. 5786, pp. 504–507, 2006. → [p7], [p9], [p22]
- [15] P. Vincent, H. Larochelle, Y. Bengio, and P.-A. Manzagol, “Extracting and composing robust features with denoising autoencoders,” in *Proceedings of the 25th International Conference on Machine Learning*, pp. 1096–1103, 2008. → [p7], [p9], [p22], [p30]
- [16] V. Chandola, A. Banerjee, and V. Kumar, “Anomaly detection: A survey,” *ACM Computing Surveys*, vol. 41, no. 3, pp. 1–58, 2009. → [p7], [p23]
- [17] P. Malhotra, A. Ramakrishnan, G. Anand, L. Vig, P. Agarwal, and G. Shroff, “Lstm-based encoder–decoder for multi-sensor anomaly detection,” in *ICML 2016 Workshop on Anomaly Detection*, 2016. → [p7]
- [18] M. Thill, W. Konen, and T. Baeck, “Temporal convolutional autoencoder for unsupervised anomaly detection in time series,” *Applied Soft Computing*, vol. 112, p. 107751, 2021. → [p7], [p9]
- [19] J.-H. Jang, T. Y. Kim, H.-S. Lim, and D. Yoon, “Unsupervised feature learning for electrocardiogram data using the convolutional variational autoencoder,” *PLOS ONE*, vol. 16, no. 12, p. e0260612, 2021. → [p7]
- [20] S. Zhou *et al.*, “Abnormal ecg detection based on an adversarial autoencoder,” *Frontiers in Physiology*, vol. 13, p. 961724, 2022. → [p7]
- [21] T. Saito and M. Rehmsmeier, “The precision–recall plot is more informative than the roc plot when evaluating binary classifiers on imbalanced datasets,” *PLOS ONE*, vol. 10, no. 3, p. e0118432, 2015. → [p8], [p9], [p26]

- [22] Y. Cui, M. Jia, T.-Y. Lin, Y. Song, and S. Belongie, "Class-balanced loss based on effective number of samples," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 9268–9277, 2019. → [p8], [p25]
- [23] C. M. Rutter, "Bootstrap estimation of diagnostic accuracy with patient-clustered data," *Academic Radiology*, vol. 7, no. 6, pp. 413–419, 2000. → [p8], [p10], [p26], [p43]
- [24] G. S. Collins, K. G. M. Moons, P. Dhiman, R. D. Riley, A. L. Beam, B. Van Calster, M. Ghassemi, X. Liu, J. B. Reitsma, M. van Smeden, A.-L. Boulesteix, J. C. Camaradou, L. A. Celi, S. Denaxas, A. K. Denniston, B. Glocker, R. M. Golub, H. Harvey, G. Heinze, M. M. Hoffman, A. P. Kengne, E. Lam, N. Lee, E. W. Loder, L. Maier-Hein, B. A. Mateen, M. D. McCradden, L. Oakden-Rayner, J. Ordish, R. Parnell, S. Rose, K. Singh, L. Wynants, and P. Logullo, "TRIPOD+AI statement: Updated guidance for reporting clinical prediction models that use regression or machine learning methods," *BMJ*, vol. 385, p. e078378, 2024. → [p8], [p10], [p58]
- [25] S. Butterworth, "On the theory of filter amplifiers," *Experimental Wireless and the Wireless Engineer*, vol. 7, pp. 536–541, 1930. → [p14], [p22]
- [26] P. Virtanen *et al.*, "Scipy 1.0: Fundamental algorithms for scientific computing in python," *Nature Methods*, vol. 17, pp. 261–272, 2020. → [p14]

A Reproducibility Record

A.1 Final experiment artifacts

The fixed temporal-holdout experiment is stored in:

- `experiments/final_temporal_holdout.json`;
- `experiments/final_temporal_holdout.joblib`; and
- `tools/final_temporal_holdout.py`.

The normal model is documented by `experiments/nsrdb_autoencoder_training.json`, produced by `tools/train_nsrdbe_autoencoder.py`. Its deployable checkpoint is `backend/app/storage/models/model_nsrdbe_primary_healthy.pt`. The deployable copy is registered by the backend model registry and stored in the backend model directory.

A.2 Frozen constants

Table A.1: Settings used for the final experiment.

Setting	Value
Experiment seed	20260801
Healthy-autoencoder seed	20260802
Healthy fitting/validation subjects	15 / 3 NSRDB records
Healthy fitting/validation windows	288,673 / 57,177
Bootstrap seed and replicates	20260801; 10,000
Development/test boundary	80% / 20% per record
Candidate training/validation	approximately 64% / 16% per record
Boundary gap	16 beats before test
Binary threshold	0.6208333333
Binary ensemble	240 fully grown Extra Trees
Subtype ensemble	260 fully grown Extra Trees
Binary/subtype input sizes	242 / 242
Independent uncertainty units	45 subject clusters

A.3 Saved success checks

The JSON artifact records three Boolean checks. Pooled whole-system accuracy above 90% is true. A whole-system subject-cluster lower 95% bound above 90% is true. Interval width at most three percentage points is false.

A.4 Safety statement

The installed artifact is a research model. The final experiment uses patient overlap, one MITDB lead, and annotated R-peak locations during evaluation. It is not a certified medical device and must not be used to diagnose or treat a patient.